Chair of Software Modeling and Verification (Computer Science 2)

RWTH AACHEN UNIVERSITY

Bachelor Thesis at the
Software Modeling and Verification Group (MOVES)

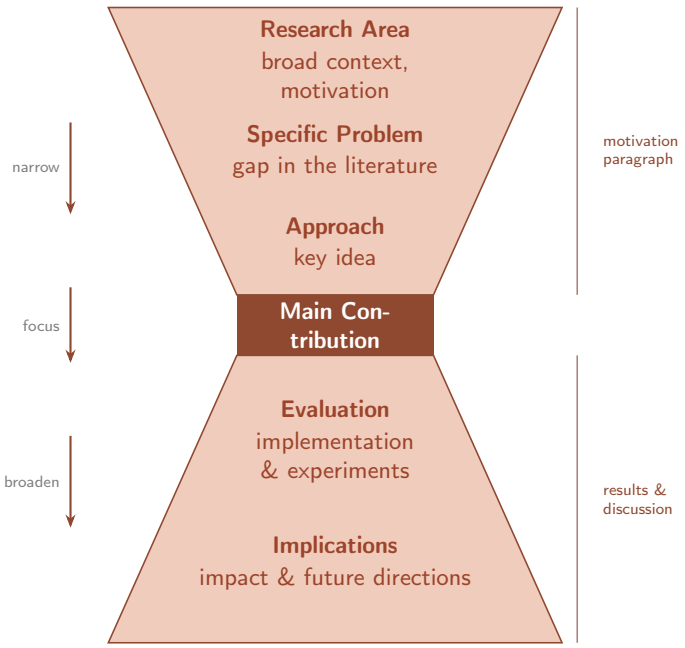
Expected Visiting Times in Interval Markov Chains

by
Jesper Nilsson

This written work was submitted to
Lehrstuhl für Informatik 2, RWTH Aachen University.

Submission Date:	September 26, 2026
First examiner:	Prof. Dr. Ir. Dr. h.c. Joost-Pieter Katoen
Second examiner:	apl. Prof. Dr. Thomas Noll
Thesis advisor:	Hannah Mertens

Abstract



Hourglass structure of a good abstract. Start broad (research area and problem), funnel down to the specific contribution, then widen again to state results and implications.

Write a self-contained summary of **roughly 150–250 words** following the hourglass structure above. Cover the following five points in order:

1. **Research area.** Name the field (e.g., strategy synthesis, probabilistic program verification, operational semantics) and state in one sentence what the field studies and why it matters.
2. **Specific problem.** State precisely which problem this thesis addresses and why it is challenging or not yet fully solved by existing approaches.
3. **Approach.** Describe the key idea of your contribution in one or two sentences. Avoid mathematical notation and abbreviations, and use easily understandable language.
4. **Theoretical results.** Name the main property proven (soundness, completeness, an equivalence result, or a complexity bound) and state it informally.

5. **Implementation and evaluation (if applicable).** Name the tool and the benchmark set and give the headline result: how many more examples are solved compared to the baseline and what is the runtime overhead.
6. **Implications.** Explain how your results solved the issue or filled the gap you pointed out beforehand.

Acknowledgements

I want to express my most profound appreciation to my supervisor!

Contents

1 Introduction **1**
- 1.1 Related Work . . . 2
- 1.2 Contribution . . . 4
- 1.3 Outline . . . 4

2 Preliminaries **5**
- 2.1 Notation . . . 5
- 2.2 Theoretical Foundation . . . 6

3 Markov Chains **11**
- 3.1 Discrete-Time Markov Chains . . . 11
- 3.2 Expected Visiting Times for DTMCs . . . 18
- 3.3 Relating Expected Visiting Times and Quantitative Reachability . . . 23

4 *Interval Markov Chains* **24**
- 4.1 Relation between the semantics . . . 35

5 *EVTs in IDTMCs* **37**
- 5.1 Calculating Expected Visiting Times of IDTMCs . . . 38
- 5.2 Relating Expected Visiting Times and Quantitative Reachability . . . 42
  - 5.2.1 UMC semantic . . . 49
  - 5.2.2 IMDP semantic . . . 50
- 5.3 P-completeness . . . 55

6 *Stationary Distributions in IDTMCs* **59**

7 Conclusion **64**

A Appendix **65**

Chapter 1

Introduction

Stochastic processes offer effective methods for analyzing real-world systems across fields like computer science, economics, and the social sciences [GT02, BLNW14, Afr20, Gal17]. One such widely used stochastic process are *discrete-time Markov chains* (DTMCs) which were introduced by Markov in 1906 [Mar06]. DTMCs consist of a set of states and a fixed transition probability function that specifies the probability of transitioning between states. They find application in a broad range of different domains, such as computer networks and software systems [GT02, PPH13], opinion dynamics [Gal17, FJ90], and biology [RC18].

When working with Markov Chains there exist multiple measures that are of interest, like reachability probabilities, stationary distributions, and expected rewards [Kat16]. Another quantity whose importance has been recognized by many authors [MKQW25, Cas09, RC18, MAK06, Afr20, GT02, SJT05, ST06, Gok04, MMAG11, PRT10, FRG13] are *expected visiting times* (EVTs), which measure how often a given model is expected to enter each state. Similarly to reachability probabilities and expected rewards, EVTs can be computed by finding the unique solution of a linear equation system [KS76].

These EVTs are used in various areas, such as determining the expected resource demands of software systems [GT02, GWHT04, SJT05, ST06], provide insights for the quantitative evaluation of reliability [Gok04, MMAG11, PRT10], and security of computer systems [OA14, AN14].

However, a fixed probability function can not always be assumed in practice [JL91, Kuz91, Wal96, KU02], for example, when the probability function is estimated through statistical experiments, which only provide bounds on the transition probabilities. To capture this imprecision of the transition probability function, *interval-valued discrete-time Markov chains* (IDTMCs) were introduce in [JL91, KU02]. These IDTMCs extend DTMCs by defining a range within which the transition probability function is assumed to lie. Various semantics have been proposed to resolve the uncertainty which arises from this imprecision. Two widely used semantics are the *once-and-for-all semantic*,

also known as the *Uncertain Markov Chain (UMC) semantic*, [JL91, SVA06] and the *IMDP semantic* [SVA06].

TODO:
Noch die semantiken stärker ausführen; kurz in einem satz erklären

As an IDTMC characterizes a set of DTMCs, the states of an IDTMC can have multiple different EVTs belonging to the contained DTMCs. We thus focus, in this thesis, on calculating the minimum and maximum EVTs for the states of an IDTMC. If we apply the aforementioned linear equation system, that can be used to calculate EVTs in DTMCs, to IDTMCs, it thus becomes an optimization problem. Moreover, the linear equation system depends on the specific transition probability function of the given DTMC. Therefore, when extending this linear equation system IDTMCs, we have to treat the transition probability function as variable as well, due to the imprecision of the transition probability function of IDTMCs. Thus, applying the linear equation system to IDTMCs to calculate the minimum and maximum EVTs necessitates solving nonlinear optimization problems. As many nonlinear optimization problems can only be solved in nondeterministic polynomial time (NP), we propose an alternative method to calculate the minimum and maximum EVTs for IDTMCs in this thesis. To our knowledge, this has not yet been studied in the literature.

1.1 Related Work

In this section, we elaborate on related work to position this thesis in the current research starting with the origins of EVTs. We continue by elaborating on topics such as expected rewards and reachability probabilities in IDTMCs.

The concept of EVTs was introduced in 1960 by Kemeny and Snell [KS76] for *absorbing DTMCs*, which consist of *absorbing* and *transient* states. The former describe states that cannot be left after they have been entered, i.e., these states only transition back to themselves. In contrast, the latter describe states that are eventually not entered again. The authors showed that the EVTs can be determined using the so-called *fundamental matrix*. In particular, the entry in the i-th row and j-th column of the fundamental matrix equals the EVT of the transient state s_j given that the unique initial state is the transient state s_i. Importantly, Kemeny and Snell also establish the notion of EVTs that coincide with the unique solution of a linear equation system.

TODO:
matrix inverse fundamental erwähnen –> um motivieren, dass (zu) aufwendig

As aforementioned, this thesis focuses on computing the minimum and maximum EVTs for the states of IDTMCs. There are multiple measures that are related to EVTs such as *expected rewards* [MKQW25] and *reachability probabilities* [BKL+17, MKQW25]. Both of these measures have been studied in IDTMCs.

As shown in [CSH08] and [SVA06], IDTMC model checking under the IMDP semantic can be reduced to model checking a Markov Decision Process (MDP). There exist many algorithms and tools to calculate expected rewards for MDPs.

TODO:
cite some examples

However, this construction results in an "exponential blowup" as [Spr18] puts it, i.e., the constructed MDP is exponential in the size of the IDTMC [CSH08].

Moreover, as shown in [BDL+17], for every IDTMC under the UMC semantic, a parametric Markov Chain can be constructed that contains the same set of DTMCs as the corresponding IDTMC. Furthermore, expected rewards for pMCs with a fixed number of parameters is in PTIME [HBK17], i.e., computing expected reward for pMCs is exponential in the number of parameters [JJK22].

TODO:
überprüfen ob wirklich das in Jansen gemeint ist

However, the construction shown in [BDL+17] creates a parameter for each interval. As we cannot bound the number of intervals of the IDTMCs without loss of generality this approach is not in PTIME.

As IDTMCs are a special case of Bounded Parameter Markov Decision Processes (BMDP), a different approach is to use the results of [GLD00, TB07] for calculating expected rewards. While [GLD00] show that expected discounted rewards for BMPDs can be calculated in PTIME, [TB07] expand this result to expected rewards.

In [Teo13] presents the calculation of expected costs, i.e., expected rewards, in IDTMCs as an optization problem. However, the objective function of this optimization problem consists of a matrix inverse. As aforementioned, the computation of an inverse becomes infeasible for large models.

Furthermore, calculating minimum and maximum reachability probabilities in IDTMCs is in PTIME for both the UMC and IMDP semantic. Moreover, these two semantics coincide on reachability probabilities [BDL+17, CHK13]. To calculate the EVT of a state, we require two different reachabilty probabilities [BKL+17]. However, an optimal reachability probability is satisfied in an IDTMC if the reachability probability is fulfilled in at least one satisfying DTMC.

TODO:
konzept satisfying schon früher einführen

When computing the minimum or maximum reachability probabilities for IDTMCs, we thus need to minimize or maximize these reachability probabilities

TODO:
vllt kurz IMRMs anschneiden; quant-analysis-semantic paper

1.2 Contribution

1.3 Outline

We begin with an overview of the notation and theoretical foundations in Chapter 2. We continue, by introducing Markov Chains, related notions, and EVTs as well as relating EVTS and quantitative reachability in Chapter 3. Furthermore, we introduce the notion of Interval Markov Chains and related notions in Chapter 4.

Thereafter in Chapter 5, we define and examine EVTs in Interval Markov Chains. To this end, we start by defining EVTs for IMCs under both semantics and provide characterizations of EVTs. We then continue, by relating EVTs with quantitative reachability and showing that the two semantics coincide on EVTs in Sec. 5.1. Then, we show how EVTs can be calculated under both the UMC and IMDP semantic in Sec. 5.2.1 and Sec. 5.2.2. Lastly, we prove P-completeness for the decisions problem of EVTs in Interval Markov Chains in Sec. 5.3. Moreover, we elaborate on how stationary distributions can be calculated in Interval Markov Chains in Chapter 6. Finally, we summarize the thesis and present possible directions for future work in Chapter 7.

Chapter 2

Preliminaries

In this chapter we introduce the theoretical foundations that are required and used throughout this thesis. We start in Sec. 2.1 by discussing the general notation used in this thesis. We then continue by introducing the theoretical foundation which is required for this thesis in Sec. 2.2.

2.1 Notation

We denote the set of natural numbers, including zero, as $\mathbb{N}$ and the set of positive natural numbers, excluding zero, as $\mathbb{N}_{>0}$. The set of real numbers is denoted as $\mathbb{R}$ and the extended real numbers by $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ for which we use the following conventions:

(i) $x + \infty = \infty + x = \infty$ for all $x \in \mathbb{R}$,

(ii) $\infty + \infty = \infty$, and

(iii) $x \cdot \infty = \infty \cdot x = \begin{cases} \infty & \text{if } x \in \mathbb{R}_{>0}, \\ 0 & \text{if } x = 0, \\ -\infty & \text{if } x \in \mathbb{R}_{<0}. \end{cases}$

TODO:
vllt noch $\mathbb{R}_{>0}$ und lz definieren?

The conventions are defined analogously for $-\infty$. Other expressions such as $\infty - \infty$ and $\frac{\infty}{\infty}$ are undefined. For a given set A we denote its powerset as $2^X = \{Y \mid Y \subseteq X\}$.

Furthermore, we denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We denote the i-th entry of a vector v as $v(i)$.

TODO:
Tue ich aber nicht immer

For convenience, we use the notation $\underline{n} = \{0, 1, 2, \ldots, n\}$ with $n \in \mathbb{N}$.

TODO:
vllt diese Notationsweise rausnehmen; und schauen ob erweiterung mit 0 passt auf alles, wo benutzt

Moreover, for a given set S we use the notation $|S|$ to denote its cardinality.

Lastly, we use *Iverson Brackets* $[\![B]\!]$ to denote the function that maps a Boolean expression B to 1 or 0 such that

$$[\![B]\!] = \begin{cases} 1 & \text{if B is true,} \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Theoretical Foundation

In this section, we introduce basic concepts from the subjects matrix, domain, measure, and probability theory which used throughout this thesis.

We start by introducing notions from domain theory. The provided definitions are taken from [MKQW25] whose material originates from [AGM95, Chapters 1].

Definition 2.1 (Partial Order)
Let X be a set with a relation $\sqsubseteq \subseteq X \times X$. The relation $\sqsubseteq$ is called a *partial order* if the following holds for all $x, y, z \in X$:

(i) $x \sqsubseteq x$ (reflexivity),

(ii) $x \sqsubseteq y$ and $y \sqsubseteq x$ implies $x = y$ (antisymmetry), and

(iii) $x \sqsubseteq y$ and $y \sqsubseteq z$ implies $x \sqsubseteq z$ (transitivity).

A partial order $\sqsubseteq$ on a set X is said to be a *linear order* if any two elements in X are comparable by $\sqsubseteq$.

We continue by briefly presenting some concepts from measure theory.

The following contents are taken from [MKQW25] whose material originates from [ADD00, Chapters 1, 4, and 5].

Definition 2.2 (σ-Algebra, Measurable Space)
Let Ω be a non-empty set. We call $\Sigma \subseteq 2^{\Omega}$ a *σ-algebra* on Ω if the following conditions are satisfied:

(i) $\Omega \in \Sigma$,

(ii) $X \in \Sigma$ implies $\Omega \backslash X \in \Sigma$ for all $X \subseteq \Omega$, and

(iii) $X_1, X_2, \cdots \in \Sigma$ implies $\bigcup_{i \in \mathbb{N}} X_i \in \Sigma$ for all $X_1, X_2, \cdots \subseteq \Omega$.

If Σ is a *sigma*-algebra on Ω, (Ω, Σ) is called a *measurable space*.

TODO:
vllt das Beispiel von Hannah mit einbauen

For the extended real numbers $\overline{\mathbb{R}}$, we consider the measurable space $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, where $\mathcal{B}(\overline{\mathbb{R}})$ is the *Borel σ-algebra* on the extended real numbers.

Definition 2.3 (Borel σ-Algebra on the Extended Real Numbers)
The Borel σ-algebra on $\overline{\mathbb{R}}$ is defined as the smallest σ-algebra $\mathcal{B}(\overline{\mathbb{R}})$ containing all intervals $(a, b] = \{x \in \overline{\mathbb{R}} \mid a < x \leq b\}$ with $a, b \in \overline{\mathbb{R}}$.

For a measurable space (Ω, Σ), a set $A \subseteq \Omega$ is called *measurable* if $A \in \Sigma$. The measurability of a function is defined as follows.

Definition 2.4 ((Borel) Measurable Function)
Let (Ω, Σ) and (Ω', Σ') be measurable spaces. A function $f : \Omega \to \Omega'$ is called *measurable* relative to Σ and Σ' if $\{\omega \in \Omega \mid f(\omega) \in f(E)\} \in \Omega$ for all $E \in \Sigma'$.

A function $f : \Omega \to \overline{\mathbb{R}}$ is called *Borel measurable* (on (Ω, Σ)) if it is measurable relative to Σ and $\mathcal{B}(\overline{\mathbb{R}})$.

Moreover, the class of Borel measurable functions is closed under addition.

Theorem 2.5 (Addition of Borel Measurable Functions)
Let (Ω, Σ) be a measurable space and let $f, g : \Omega \to \overline{\mathbb{R}}$ be two Borel measurable functions. The function $f + g$, defined by pointwise addition, is measurable if it is well-defined, i.e., if $f(\omega) = \infty$ implies $g(\omega) \neq -\infty$ and $f(\omega) = -\infty$ implies $g(\omega) \neq \infty$ for all $\omega \in \Omega$.

Additionally, to discuss the probability of measurable sets, we introduce the notion of *probability measures*.

Definition 2.6 (Probability Measure, Probability Space)
Let (Ω, Σ) be a measurable space. A function $\Pr : \Sigma \to [0, 1]$ is a *probability measure* on (Ω, Σ) if $\Pr(\Omega) = 1$ and for every countable collection $A_1, A_2, \ldots \in \Sigma$

of disjoint sets it holds that

$$\Pr\left(\bigcup_{k\geq 1} A_k\right) = \sum_{k\geq 1} \Pr(A_k).$$

If $\Pr : \Sigma \to [0,1]$ is a probability measure on (Ω, Σ), then $(\Omega, \Sigma, \Pr)$ is called a *probability space*.

In a probability space $(\Omega, \Sigma, \Pr)$, the elements of Ω are called outcomes, while the measurable subsets of Ω are referred to as events. An event $E \in \Sigma$ is said to hold *almost surely* if $\Pr(E) = 1$.

TODO:
Lebesgue Integral; einf. ähnlich wie Hannah oder Def. aus Ash, doleans-dade nehmen

In this thesis, we make use of the Lebesgue integral $\int_\Omega f(\omega) dPr(\omega)$ of a Borel measurable function $f : \Omega \to \overline{\mathbb{R}}$ and a probabiltiy measure Pr on a probability space (Ω, Σ). We presuppose that the reader is familiar with the notion of Lebesgue integration. A thorough examination can be found in [ADD00, Chapter 1.5].

Definition 2.7 (Expected Values)
Let (Ω, Σ, Pr) be a probability space. If $f : \Omega \to \overline{\mathbb{R}}$ is a Borel measurable function on (Ω, Σ, Pr), then the *expected value* of f, denoted as $\mathbb{E}[f]$, is the Lebesgue integral of f with respect to the probability measure Pr, i.e., $\mathbb{E}[f] = \int_\Omega f(\omega) dPr(\omega)$, provided the integral is well-defined.

TODO:
von Hannah: Throughout this thesis, we only consider the Lebesgue integral of non-negative Borel measurable functions, which is always well-defined and can be infinity.

A *probability distribution* is a function that assigns probabilities to outcomes.

Definition 2.8 (Discrete Probability Distribution)
A function P is a *discrete probability distribution* with $P : X \to [0,1]$, where X is a countable set and $\sum_{x \in X} P(x) = 1$.

TODO:
Brauche ich cond. prob. und law of total prob??

We continue by introducing notions from matrix theory based on [**?**].

An important notion, which we require in this thesis, are *(strictly) row-stochastic ma-*

trices.

Definition 2.9 (Row-Stochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$.

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions.

Definition 2.10 ((Strictly) Row-Substochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-substochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} \leq 1$ for all rows $i \in \underline{n}$.
$\mathcal{T}$ is strictly row-substochastic if and only if ther exists at least one row i' with $\sum_{j=0}^{n} (\mathcal{T})_{i'j} < 1$

If a matrix A converges to the null matrix, then the inverse of $(\boldsymbol{I} - A)$, where $\boldsymbol{I}$ is the identity matrix, is equal to the so-called Neumann series [Neu77].

Definition 2.11 (Matrix inversion by the Neumann Series)
Given a matrix $A \in \mathbb{R}^{n \times n}$ with $\lim_{k \to \infty} A^k = 0$, i.e. A is convergent, the Neumann series $\sum_{k=0}^{\infty} A^k$ converges against $(\boldsymbol{I} - A)^{-1}$.

TODO:
benutze ich full rank?

We will now shortly introduce linear and nonlinear programming based on [LY08].

Definition 2.12 (Linear Programming)
A *linear programming* problem with n variables and m constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & c^{\mathsf{T}} x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \ , \end{aligned}$$

where $c, x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.

In contrast to a linear programming, a nonlinear programming has a nonlinear objective function or nonlinear constraints.

Definition 2.13 (Nonlinear Programming)
A *nonlinear programming* problem with n variables and $m + r$ constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & f(x) & \\ \text{s.t.} \quad & g_i(x) \leq 0 & \quad i \in \underline{m} \\ & h_j(x) = 0 & \quad j \in \underline{r} \\ & x \in S \ , & \end{aligned}$$

where $x \in \mathbb{R}^n$ and $f, g_i, h_j : S^n \to \mathbb{R}$ for all $i \in \underline{m}$, $j \in \underline{r}$. At least one of the functions f, g_i, or h_j is nonlinear.

Lastly, we use the convergence of the geometric series in this thesis.

Definition 2.14 (Convergence of Geometric Series)
The geometric series $\sum_{k=0}^{\infty} q^k$ converges for all $q \in \mathbb{R}$ with $|q| < 1$, where

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \ , \quad \text{for } |q| < 1$$

TODO:
include geo. series?; if yes cite it

Chapter 3

Markov Chains

The previous chapter provided preliminary mathematical notions. In the following we introduce the notion of Markov Chains, which are central to this thesis.

This chapter starts with a formal definition of Markov Chains in Sec. 3.1. We then continue, by elaborating on related concepts, such as the probability space, reachability probabilities, as well as recurrent and transient states. Moreover, we describe a further important concept, the *expected visiting times* of states in DTMCs, in Sec. 3.2 which were first introduced by [KS76]. Lastly, we relate these *expected visiting times* to reachability probabilities in Sec. 3.3, as done in [BKL+17, MKQW25].

3.1 Discrete-Time Markov Chains

In this section, we will introduce definitions and theorems for Markov Chains. The notation and contents of this section are mostly based on [BK08], [MKQW25], and [CSH08].

Definition 3.1 (DTMC)
A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota_{init}^{\mathcal{D}})$, where

(i) S is a countable, nonempty set of states,

(ii) $\mathbf{P} : S \times S \to [0,1]$ is the *transition probability function*, with $\sum_{s' \in S} \mathbf{P}(s,s') = 1$ for all states $s \in S$, and

(iii) $\iota_{init} : S \to [0,1]$ is the *initial distribution*, such that $\sum_{s \in S} \iota_{init}(s) = 1$.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $S^{\mathcal{D}}$, $\mathbf{P}^{\mathcal{D}}$, and $\iota_{init}^{\mathcal{D}}$. We assume in this thesis, that the DTMCs are *finite*, i.e., that the set of states S is *finite*. We sometimes refer to these finite DTMCs, as *finite-state DTMCs*. We call a state $s \in S$ is called *initial* if it satisfies $\iota_{init}(s) > 0$. Furthermore, we assume

an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s_i, s_j \in S$ and $i < j$. This allows us, for example, to denote the i-th entry of the initial distribution as $\iota_{init}(s_i)$, when viewing ι_{init} as a column vector. Due to the conditions (i) and (ii) provided in Def. 3.1, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions, see Def. 2.8.

TODO:
discrete prob. distr. comment necessary?

TODO:
introduce concept of a fixed linear order in Matheteil, probablby can just copy Hannah

TODO:
??(notion of a current state)

Moreover, if the probability to transition from a state s_i to s_j, for two given states $s_i, s_j \in S$, is greater than zero, i.e., $\mathbf{P}(s_i, s_j) > 0$, then we refer to s_j as a direct successor of s_i. Conversely, we refer to s_i as a direct predecessor of s_j. Additionally, we define a neighbour set N, with $s_j \in N(s_i)$, if and only if s_j is a direct successor of s_i.

Paths and Probability Space. To reason about the probabilistic behavior of a DTMC $\mathcal{D}$, we define its probability space, in which the outcomes correspond to the infinite paths of $\mathcal{D}$.

To this end, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ for the rest of this section.

Definition 3.2 (Path of a DTMC)
An *infinite path* π in a DTMC $\mathcal{D}$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$. A *finite path* is a finite prefix of an *infinite path*.

We use the notation $\pi[k] \in S$ to denote the k-th element of the path π for all $k \in \mathbb{N}_0$, i.e. $\pi[k] = s_k$.

Notice, that for any path π a given state $\pi[k]$, which is visited in step $k \in \mathbb{N}$, is always followed by one of its direct succesors, i.e., $\pi[k+1] \in N(\pi[k])$. Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{D}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path is infinite in lenght, we define $\pi \in Paths^{\mathcal{D}}_{inf}$ to be $|\pi| = \infty$. We denote the last state of a finite path $\pi \in Paths^{\mathcal{D}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$. The sets of all infinite and finite sets are denoted as $Paths^{\mathcal{D}}_{inf}$ and $Paths^{\mathcal{D}}_{fin}$, respectively.

Further, we construct the *probability space* of a DTMC as follows. The events corre-

spond to sets of infinite paths that can be expressed as countable unions and intersections of cylinder sets, i.e., sets containing all possible continuations of finite paths in $\mathcal{D}$.

TODO:
cylinder set sachen, die hannah reingepackt hat bei sich?; cylinder sets in matheteil einführen

Definition 3.3 (Probability Space of a DTMC)
The *probability space* of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$ for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

  TODO:
  Define σ-algebra, bei Hannah Def. 2.12 ff

- $Pr^{\mathcal{D}}$ is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

  for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$

TODO:
define σ-algebra in Matheteil

Additionally, we use the notation $Pr^{\mathcal{D}}_s$ to denote the probability measure on the measurable space induced by fixing $s \in S$ as the unique initial state. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $Pr^{\mathcal{D}}$ and $Pr^{\mathcal{D}}_s$.

TODO:
measurable space definieren? Macht Hannah das?

Reachability probability. In the following, we consider events, i.e., measurable sets of infinite paths, which are relevant for the introduction of reachability probabilities in DTMCs.

TODO:
measurable sets in Matheteil einführen

First, we define the *until* operator (U), see [BK08].

Definition 3.4 (Until Operator)
Let $R, T \subseteq S^{\mathcal{D}}$ be two sets of states of a DTMC $\mathcal{D}$. For these two sets, we define the *until* operator as

$$R \,\mathsf{U}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Futhermore, the *step-bounded until* operator is defined as

$$R \,\mathsf{U}^{=k}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Intuitively, the set $R \,\mathsf{U}\, T$ contains all paths that only visit states in R, until a state in T is visited.

Using the *until* operator, we can introduce the *eventually* operator ($\Diamond$).

Definition 3.5 (Eventually Operator)
Let $T \subseteq S$ be a set of states of a DTMC $\mathcal{D}$. Then we define the *eventually* operator as $\Diamond T = S \,\mathsf{U}\, T$.

Consequently, we define the *step-bounded eventually* operator as $\Diamond^{=k} T = S \,\mathsf{U}^{=k}\, T$.

In other words, $\Diamond T$ contains all paths $\pi \in Paths^{\mathcal{D}}_{fin}$ that visit at least one $t \in T$ at some point, i.e., there exists at least one $k \in \mathbb{N}$ with $k \leq |\pi|$, such that $\pi[k] = t$.

With the *eventually* operator, we can define the *reachability probability*.

Definition 3.6 (Reachability Probability)
The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$.

TODO:
eigener Definitionsblock hierfür?

Thus, the reachability probability describes the probability of visiting a set of states in a DTMC $\mathcal{D}$. Intuitively, if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable in $\mathcal{D}$, i.e., there exists at least one $\pi \in Paths^{\mathcal{D}}_{fin}$, such that $\pi[0]$ is an initial state of $\mathcal{D}$ and π visits t. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$). Lastly, we denote $\Diamond_t T = \{\pi \in \Diamond T \mid \pi[0] = t\}$.

To calculate the reachability probability in this thesis, we often use the recursive formula, found in the following lemma.

Lemma 3.7 (Recursive Formula for Reachability Probability)
For states $t \in S$ and $T \subseteq S$:

$$Pr_t(\Diamond T) = \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)$$

with $Pr_t(\Diamond T) = 1$ for all $t \in T$.

We will provide a short proof for this lemma.

Proof 1
Using Def. 3.3,

$$\begin{aligned}
Pr_t^{\mathcal{D}}(\Diamond T) &= \int_{\pi \in \Diamond T} \iota_{init}^{\mathcal{D}}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \int_{\pi \in \Diamond_t T} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as t is the unique initial state, i.e., } \iota_{init}(t) = 1.) \\
&= \sum_{u \in N(t)} \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as } t \text{ can only be succeded by } u \in N(t)) \\
&= \sum_{u \in N(t)} \mathbf{P}(t,u) \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(with } \pi[0] = t, \pi[1] = u)
\end{aligned}$$

For $\pi \in \Diamond_t T$ with $\pi[1] = u$, define $\pi' = (\pi[1], \ldots, last(\pi))$. As $\pi'[0] = u$ and $last(\pi') = s$, $\pi' \in \Diamond_u T$. Conversely, for $\pi' \in \Diamond_u T$, define $\pi = (t, \pi'[0], \ldots, last(\pi'))$, then $\pi \in \Diamond_t T$. Thus,

$$\begin{aligned}
&= \sum_{u \in N(t)} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) \\
&= \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) && \text{(as, if } u \notin I_{tT}, \text{ then } Pr_u(\Diamond T) = 0, \text{ see Def. 5.7.)} \\
& && (3.1)
\end{aligned}$$

□

TODO:
maybe to much explanation & reicht das so?

TODO:
doppel pi sehr hässlich, aber für last(pi) die dots notation nicht genauso klar

Lastly, we use the *next* operator ($\bigcirc$), which is defined as follows.

Definition 3.8 (Next Operator)
Let $\mathcal{D}$ be a DTMC and let $E \in \sum_{Paths^{\mathcal{D}}_{inf}}$ be a measurable set of infinite paths in $\mathcal{D}$. We now define the *next* operator, as

$$\bigcirc E = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid (\pi[1], \pi[2], \dots) \in E\} \ ,$$

TODO:
define measurable sets

TODO:
mention, that operators in LTL-style notation?

In other words, a path $\pi \in Paths^{\mathcal{D}}_{inf}$ is contained in $\bigcirc E$ if and only if $\pi' = (\pi[1], \pi[2], \dots)$ is contained in E. Moreover, the next step probability is defined as $Pr(\bigcirc E) = \sum_{s' \in N(s)} \mathbf{P}(s, s') \cdot Pr_{s'}(E)$.

In this thesis, we use the *next* operator ($\bigcirc$) in combination with the *eventually* operator ($\Diamond$). Specifically, we use them to denote the *recurrence probability* $Pr_s(\bigcirc\Diamond s)$, i.e., the probability to reach s from all of its successors.

Strongly Connected Components. We can partition the state set of a DTMC as shown in the following definition.

Definition 3.9 ((Bottom) Strongly Connected Component)
A non-empty set of states $C \subseteq S^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is *strongly connected* if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths^{\mathcal{D}}_{inf}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset of C, i.e. $C' \subset C$, is strongly connected, then C is called a *strongly connected component* (SCC).

Furthermore, an SCC C is called a *bottom strongly connected component* (BSCC) if there exists no $k \in \mathbb{N}$ and no path $(\pi[0], \dots, \pi[k], \dots) \in Paths^{\mathcal{D}}_{inf}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once a state of a BSCC C is entered, C cannot be left again.

Using BSCCs, we can define *acyclic* DTMCs.

Definition 3.10 (Acyclic DTMC)
A DTMC $\mathcal{D}$ is called *acyclic* if and only if each SCC is either a BSSC or a singleton set $\{s\}$ with $\mathbf{P}(s,s) = 0$

Recurrent and Transient States. Using the *recurrence probability* $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s)$, we can partition the state set into *recurrent* and *transient* states.

Definition 3.11 (Recurrent and Transient States)
For a given DTMC $\mathcal{D}$ and state $s \in S$, s is called *recurrent* if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not *recurrent* are called *transient*.

In other words, a state is *recurrent* precisely, when we are guaranteed to reach it from all of its direct successors, meaning once we reach a *recurrent* state, we are bound to return to it. Using this definition, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states. Notice, that *recurrent* states are precisely the states contained in a BSCC. For convenience, we additionally define the set of *initial* states $S_{in}^{\mathcal{D}} = \{s \mid \iota_{init}^{\mathcal{D}}(s) > 0\}$. Furthermore, we define the set of *transient initial* states $S_{tr \cap in} \subseteq S_{in}$ with $S_{tr \cap in} = S_{in} \cap S_{tr}$.

Moreover, the event $\Diamond S_{re}$, i.e., to reach any recurrent state, holds *almost surely* in any DTMC.

TODO:
Explain almost surely

Theorem 3.12 (Limit Behaviour)
For any DTMC $\mathcal{D}$, set of recurrent states S_{re}, and any state $s \in S$, it holds that $\lim_{k \to \infty} Pr_s(\Diamond^{=k} S_{re}) = 1$.

Intuitively, the probability of reaching a recurrent state after k steps, starting in s, tends towards 1 as k tends towards infinity.

Lastly, we define *absorbing* states.

Definition 3.13 (Absorbing state)
A states s of a DTMC $\mathcal{D}$ is refered to as an *absorbing state* if and only if $\mathbf{P}^{\mathcal{D}}(s,s) = 1$.

This means, once the model enters an *absorbing* state, it cannot leave it. Notice, that *absorbing* states are a special type of *recurrent* states, as the only direct successor of an *absorbing* state $s \in S$ is s itself and therefore trivially $Pr_s(\Diamond s) = 1$. Thus, we are guaranteed to return to a given *absorbing* state after one step.

Definition 3.14 (Markov Descision Process (MDP))
[BK08]

TODO:
größtenteils übernommen

A *Markov Decision Process* (MDP) is a tuple $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$, where

(i) S is a countable set of states

(ii) Act is a set of actions

(iii) $A(s) \subseteq Act$ denotes the applicable actions for all states $s \in S$

(iv) $\mathbf{P} : S \times Act \times S \to [0, 1]$ is the transition probability function such that $\sum_{s' \in S} \mathbf{P}(s, \alpha, s') = 1$ for all $s \in S$ and $\alpha \in A(s)$

(v) $\iota_{init} : S \to [0, 1]$ is the initial distribution such that $\sum_{s \in S} \iota_{init}(s) = 1$

Definition 3.15 (Canonical Form)
For a DTMC $\mathcal{D}$, its transition matrix $\mathcal{T}$ is said to be in canonical form if and only if

$$\mathcal{T} = \begin{bmatrix} Q & R \\ 0 & S \end{bmatrix}$$

where

(i) $Q = (\mathbf{P}(s,t))_{s,t \in S_{tr}}$ contains the transition probabilities between the transient states

(ii) $R = (\mathbf{P}(s,t))_{s \in S_{tr}, t \in S_{re}}$ contains the transition probabilities between transient and recurrent state pairs

(iii) $S = (\mathbf{P}(s,t))_{s,t \in S_{re}}$ contains the transition probabilities between the recurrent states

For absorbing DTMCs $S = \boldsymbol{I}$. As mentioned in **??**, this is precisely possible for

TODO:
remove canonical form (as I do not use it); replace it with definition of Q, if not already have that somewhere

3.2 Expected Visiting Times for DTMCs

In this section, we will introduce the concept of *expected visiting times*, which were first introduced by Kemeny and Snell [KS76] in the context of the *fundamental matrix*.

TODO:
vllt noch mehr zu was vorgestellt; wie bei Hannah: fundamental aufwendig weil ..., deshalb ...

The notation and contents of this section are mostly based on [MKQW25] and [KS76].

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ and start by introducing *visiting times* along paths.

Definition 3.16 (Visiting Times of Infinite Paths)
The function $vt_s : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ describes the *visiting time* of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] \ .$$

Similarly, the function $vt_T : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ descsribes the *visiting time* of a set of states $T \subseteq S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$

$$vt_T(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \ .$$

As proven in [MKQW25], the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S^{\mathcal{D}}$. Using this result, we can shortly prove the same for vt_T.

Proof 2
Using def. 3.16,

$$\begin{aligned} vt_T(\pi) &= \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \\ &= \sum_{k=0}^{\infty} \sum_{t \in T} [\![\pi[k] = t]\!] \\ &= \sum_{t \in T} vt_t(\pi) \ . \end{aligned}$$

Thus, the Lebesgue integral of the non-negative function vt_T is well-defined for all sets of states $T \subseteq S$. □

TODO:
vllt das lemma/whatever inkludieren, dass ich die summen swappen darf

With this function, we can now define the EVT of a state s as the expected value of vt_s over all possible paths.

Definition 3.17 (Expected Visiting Times in DTMCs)
For a state $s \in S$, the *expected visiting time* of s is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the *expected visiting time* that is obtained by fixing a certain state $t \in S$ as the only initial state as $\mathbb{E}^{\mathcal{D}}_t[vt_s]$. It is defined as

$$\mathbb{E}^{\mathcal{D}}_t[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}_t(\pi) \ .$$

Lastly, for a set of states $T \subseteq S$ the *expected visiting time* of T is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_T] = \sum_{t \in T} \mathbb{E}[vt_t] \ .$$

Because the probability measure of paths π, see definition Def. 3.3, is defined as a product containing the factor $\iota_{init}(\pi[0])$, it is nonzero only for paths that begin in a state s with $\iota_{init}(s) > 0$. Thus, the value $\mathbb{E}^{\mathcal{D}}[vt_s]$ gives the number of times the state s is expected to be visited from all paths in $\mathcal{D}$ that start in one of the initial states. Whenever the DTMC $\mathcal{D}$ is evident from the context, we will omit the superscript $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$, $\mathbb{E}^{\mathcal{D}}_t[vt_s]$, and $\mathbb{E}^{\mathcal{D}}[vt_T]$.

We shortly prove the third equation of Def. 3.17.

Proof 3

$$\begin{aligned}
\mathbb{E}[vt_T] &= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_T(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{t \in T} vt_t(\pi) dPr^{\mathcal{D}}(\pi) && \text{(using Def. 3.16)} \\
&= \sum_{t \in T} \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_t(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \sum_{t \in T} \mathbb{E}[vt_t] \ . && \text{(using Def. 3.17)} \qquad \square
\end{aligned}$$

TODO:
bisschen text hinzufügen

TODO:
diese ganze T geschichte für EVTs natürlich auch eine eigene contribution

Using the aforementioned partition of the set of states S into transient and recurrent states, see Def. 3.11, we can already give a characterization for the latter.

Lemma 3.18 (EVTs of Recurrent States)
For all recurrent states $s \in S_{re}$, it holds that

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if s is reachable,} \\ 0 & \text{else.} \end{cases}$$

For a proof of this lemma, see [MKQW25].

Thus, we will focus on finding the non-trivial EVTs of the transient states. For this, we will first define the *fundamental matrix* which was introduced by Kemeny and Snell [KS76].

Definition 3.19 (Fundamental Matrix)
For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states (defined in Def. 3.15), the *fundamental matrix* F is defined as,

$$F = (I - Q)^{-1} \ .$$

Kemeny and Snell show in [KS76] that Q converges to the null matrix. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty} Q^k = (\boldsymbol{I} - Q)^{-1}$, see Def. 2.11, i.e., the fundamental matrix is well-defined.

Importantly for this thesis, Kemeny and Snell show in [KS76], that the fundamental matrix gives for any two transient states $s, t \in S_{tr}$ the expected number of times s is visited, when starting in t. To be more precise, $\mathbb{E}_t[vt_s] = (F)_{ts}$. From this equation, it intuitively follows that transient states are only visited a finite amount of times.

TODO:
for this equation to hold, F needs to be ordered. Just mention this shortly

TODO:
make reach of tran states into remark? –> if yes: rework the proof

TODO:
ist der proof so sauber/schön genug + soll das ein lemma oder proof sein? + notion eines current state einführen?; falls ich das niergendswo mehr benutze, könnte ich es trotzdem als remark oder so verwenden

Thus, we can use the fundamental matrix to characterize the EVTs for transient states. However, as the calculation of the inverse of a high-dimensional matrix becomes infeasible for large models, we will take a look at how the EVTs can be calculated using linear equation systems, as shown in [MKQW25].

TODO:
zweiter satz fast eins zu eins von Hannah übernommen; vllt noch abändern

First, we define

$$\tau_{init} = (\iota_{init}(s_1), \ldots, \iota_{init}(s_n))^{\mathsf{T}}$$

for all $s \in S_{tr}$ with which we can then introduce the following theorem taken from [MKQW25].

Theorem 3.20 (Linear Equation System Characterization of EVTs of Transient States) The vector $(\mathbb{E}[vt_s])_{s \in S_{tr}}$ provides the unique solution of the following linear equation system in $\mathbb{R}^{|S_{tr}|}$:

$$(\boldsymbol{I} - Q^T) \cdot x = \tau_{init} \ .$$

TODO:
Für Matheteil: $\boldsymbol{I}$ denotes identity matrix

For a proof of this theorem, see [MKQW25].

This equation system can then be rewritten as

$$x = \tau_{init} + Q^{\mathsf{T}} \cdot x \ .$$

For a single state $s \in S_{tr}$, we can thus calculate the respective EVT as follows,

$$\mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s) \cdot \mathbb{E}[vt_t] \ . \tag{3.2}$$

3.3 Relating Expected Visiting Times and Quantitative Reachability

We now relate *expected visiting times* to *quantitave reachability* for DTMCs. This has already been done in [BKL+17, Lemma 3.5] and [MKQW25, Lemma 5.1]. Thus, we present their results in this section.

Lemma 3.21
Given a DTMC $\mathcal{D}$ and state $s \in S^{\mathcal{D}}_{tr}$, the EVT of s is given by

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)} \ .$$

In other words, we can calculate the EVT of a DTMC $\mathcal{D}$ with only two reachability probabilities. For a proof of Lemma 3.21, see [BKL+17, Lemma 3.5] and [MKQW25, Lemma 5.1].

Chapter 4

Interval Markov Chains

The focus of this thesis are expected visiting times in Interval Markov Chains, which we will introduce in this chapter. In the following sections we will mainly discuss how the expected visiting times can be calculated and shortly examine stationary distributions of Interval Markov Chains.

TODO:
Shortly motivate what an Interval Markov Chain is; use introduction and related works to do this

We start by introducing concepts which we will use in the following sections.

Definition 4.1 (Interval Markov Chain)
An interval discrete-time Markov chain (IDTMC) is a tuple $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$, where

(i) $S^{\mathcal{I}}$ is a countable, nonempty set of states,

(ii) $\underline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the lower bound on the transition probability function,

(iii) $\overline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the upper bound on the transition probability function,

(iv) $\underline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the lower bound on the initial distribution, and

(v) $\overline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the upper bound on the initial distribution,

such that $\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$ and $\underline{\iota}^{\mathcal{I}}_{init}(s) \leq \overline{\iota}^{\mathcal{I}}_{init}(s)$ for all $s \in S^{\mathcal{I}}$.

TODO:
for all statements in eins zusammenfassen?

TODO:
was sind die PFs und iotas hier? Satisfying DTMC als Konzept noch nicht eingeführt; stattdessen einfach nur PFlower <= PFupper etc.?

Whenever the IDTMC $\mathcal{I}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{I}}$, and $\iota_{init}^{\mathcal{I}}$.

In contrast to the precise transition probability function and initial distribution of DTMCs, Interval Markov Chains allow for uncertainty.

Example 4.2
The Fig. 4.1 depicts an exemplary IDTMC $\mathcal{I}_E = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, where $S = \{s_1, s_2, s_3, s_4, s_5\}$, the lower and upper transition probability functions are given by

$$\underline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.2 \\ 0.2 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \overline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the lower and upper initial distributions

$$\underline{\iota}_{init} = (0.5, 0, 0, 0, 0)^{\mathsf{T}} \qquad \overline{\iota}_{init} = (1, 0, 0.5, 0, 0)^{\mathsf{T}}.$$

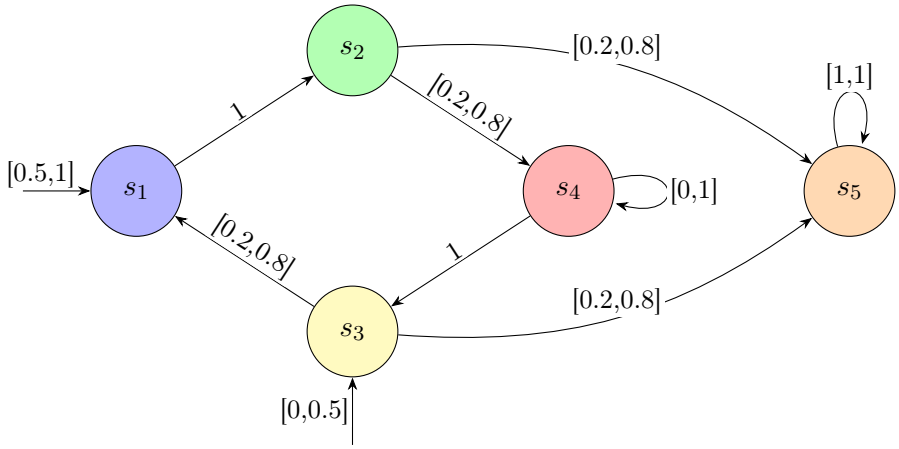


Figure 4.1: IDTMC $\mathcal{I}$

TODO:
grafik erweitern, sodass auch init ein Intervall hat. Damit "sinnvoll" zweiten init state; muss dann auch entsprechend die Semantik Grafiken anpassen

TODO:
sollen diese Grafiken wirklich so farbig sein?

Definition 4.3 (Path of an IDTMC)
Similarly to DTMCs, an *infinite path* in an IDTMC $\mathcal{I} = (S,)$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S^{\mathcal{I}}$, such that $\underline{\mathbf{P}}(\pi[k], \pi[k+1]) > 0$ for all $k \in \mathbb{N}$.

Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{I}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path has infinite length, we define $\pi \in Paths^{\mathcal{I}}_{inf}$ to be $|\pi| = \infty$. Moreover, we denote the last state of a finite path $\pi \in Paths^{\mathcal{I}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$.

Fixed-initial-state definition. In the literature, see for example [CHK13, SVA06, BDLM21], it is common to represent the initial distribution by an additional initial state rather than introducing a separate initial-distribution function. In this case, an IMC is given as a tuple $\mathcal{I}_{\text{fix}} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, s_{\text{in}})$, where s_{in} is the unique initial state. We refer to this as the *fixed-initial-state definition*.

More precisely, let $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$ be an IMC in our definition. The corresponding *fixed-initial-state* representation is obtained by introducing a fresh state s_{in} and defining $S^{\mathcal{I}_{\text{fix}}} = S^{\mathcal{I}} \cup \{s_{\text{in}}\}$.

The lower and upper transition probability functions are given by

$$\underline{\mathbf{P}}^{\mathcal{I}_{\text{fix}}}(s_i, s_j) = \begin{cases} \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) & \text{if } s_i, s_j \neq s_{\text{in}}, \\ \underline{\iota}^{\mathcal{I}}_{init}(s_j) & \text{if } s_i = s_{\text{in}}, \\ 0 & \text{otherwise} \end{cases}$$

and

$$\overline{\mathbf{P}}^{\mathcal{I}_{\text{fix}}}(s_i, s_j) = \begin{cases} \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) & \text{if } s_i, s_j \neq s_{\text{in}}, \\ \overline{\iota}^{\mathcal{I}}_{init}(s_j) & \text{if } s_i = s_{\text{in}}, \\ 0 & \text{otherwise.} \end{cases}$$

The same construction applies to DTMCs. Let $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota^{\mathcal{D}}_{\text{init}})$ be a DTMC. Its corresponding *fixed-initial-state* representation is $\mathcal{D}_{\text{fix}} = (S^{\mathcal{D}} \cup \{s_{\text{in}}\}, \mathbf{P}^{\mathcal{D}_{\text{fix}}}, s_{\text{in}})$, where

$$\mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_i, s_j) = \begin{cases} \mathbf{P}^{\mathcal{D}}(s_i, s_j) & \text{if } s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{D}}_{\text{init}}(s_j) & \text{if } s_i = s_{\text{in}}, \\ 0 & \text{otherwise.} \end{cases}$$

The two representations induce the same probability measure under the natural bijection between their paths. In particular, we define

$$\phi : Paths^{\mathcal{D}_{\text{fix}}}_{\text{inf}} \longrightarrow Paths^{\mathcal{D}}_{\text{inf}}, (s_{\text{in}}, s_0, s_1, \dots) \mapsto (s_0, s_1, \dots) \ .$$

Notice, that ϕ is a bijection, since s_{in} has no incoming transitions, every infinite path of $\mathcal{D}_{\text{fix}}$ starts with s_{in}.

To see that the probability measures correspond under ϕ, consider a finite path $\pi_{\text{fix}} = (s_{\text{in}}, s_0, \ldots, s_n)$ of $\mathcal{D}_{\text{fix}}$, and let $\pi = (s_0, \ldots, s_n)$ be the corresponding finite path of $\mathcal{D}$. Then

$$Pr^{\mathcal{D}_{\text{fix}}}(\pi_{\text{fix}}) = \mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_{\text{in}}, s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}_{\text{fix}}}(s_k, s_{k+1}) \quad = \iota^{\mathcal{D}}_{\text{init}}(s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}}(s_k, s_{k+1}) = Pr^{\mathcal{D}}(\pi).$$

Thus, corresponding cylinder sets have the same probability, and consequently the probability measures on the two path spaces coincide under the bijection ϕ.

Hence, adding a unique initial state is merely a different representation of the initial distribution and does not change the probabilistic behaviour of the model. In particular, results formulated for the *fixed-initial-state definition* can be transferred to our definition by applying the above construction.

To resolve the uncertainty in the initial distribution and transition probability function, we need to define semantics that specify a fixed initial distribution and transition probability function. In other words, the semantics specify a set of Markov Chains that are consistent with the given Interval Markov Chain. We say that all of these DTMCs $\mathcal{D}$ satisfy the IDTMC $\mathcal{I}$ and denote it as $\mathcal{D} \models \mathcal{I}$. We assume in this thesis that there exists at least one satisfying DTMC for every IDTMC.

Assumption 1
In this thesis, we solely consider finite IDTMCs $\mathcal{I}$, i.e., $S^{\mathcal{I}}$ is finite. We sometimes refer to these finite IDTMCs, as *finite-state IDTMCs*. Moreover, we restrict our consideration to IDTMCs for which, for every state $s \in S^{\mathcal{I}}$, there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$, such that s is reachable with a positive probability from an initial state in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$. If a given state were unreachable in every satisfying DTMC, then its EVT would trivially be 0 for all satisfying DTMCs and, consequently, could not contribute to the EVTs of other states.

In this thesis, we consider two popular semantics: the *once-and-for-all semantic*, also known as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic*. We will first define the former, similar to [BDLM21, CSH08, BLW13]. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$ for the rest of this chapter.

Definition 4.4 (Once-and-for-all/UMC Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ under the UMC semantic if and only if $S^{\mathcal{I}} = S^{\mathcal{D}}$ and

$$\mathbf{P}^{\mathcal{D}}(s_i, s_j) \in [\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)] \qquad \text{and} \qquad \iota^{\mathcal{D}}_{init}(s) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$$

for all $s_i, s_j, s \in S^{\mathcal{I}}$.

In other words, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC, is satisfactory. Thus, we can understand the UMC semantic for IDTMCs, as choosing the initial distribution and probability function from the intervals *once-and-for-all*, hence its name. If $\mathcal{D}$ satisfies $\mathcal{I}$ under the UMC semantic, we denote this as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$. As $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$ implies $S^{\mathcal{D}} = S^{\mathcal{I}}$, we use the same symbol $s \in S^{\mathcal{D}} = S^{\mathcal{I}}$ to denote the corresponding state in both $\mathcal{D}$ and $\mathcal{I}$.

Example 4.5
Consider the IDTMC $\mathcal{I}_E$ in Fig. 4.1. The DTMC $\mathcal{D}_U = (S^{\mathcal{I}_E}, \mathbf{P}, \iota_{init})$, depicted in Fig. 4.2, satisfies $\mathcal{I}_E$ under the UMC semantic, i.e., $\mathcal{D}_U \models_U \mathcal{I}_E$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution

$$\iota_{init} = (1, 0, 0, 0, 0) .$$

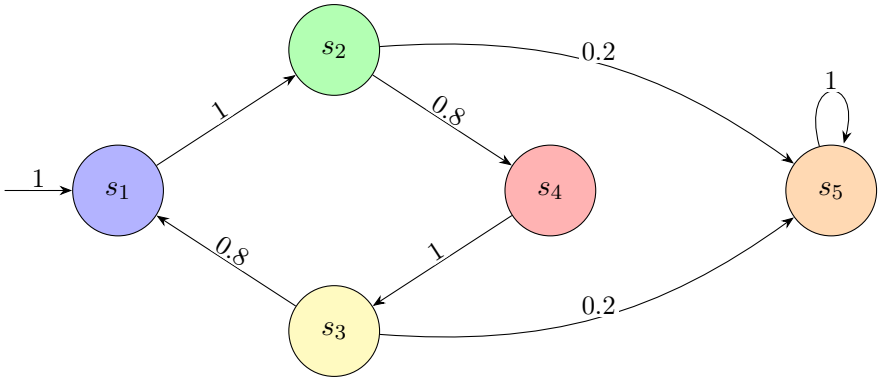


Figure 4.2: DTMC $\mathcal{D}_U$ with $\mathcal{D}_U \models_U \mathcal{I}$

The second semantic which we will define, is the *IMDP semantic*. There are two common definitions for this semantic. In this thesis, we will use the definition found in [SVA06] and [CSH08].

TODO:
Hannah: add intuitive explanation; reicht das was jetzt habe?

While the *UMC* semantic selects a particular probability function *once-and-for-all*, the *IMDP* semantic allows for a probability function to be picked from the set of valid functions before every transition.

TODO:
"valid" functions?; gute Formulierung?

Similar to [SVA06] and [CSH08], we define the set of valid probability functions over $S^{\mathcal{I}}$ as $Steps^{\mathcal{I}}(s)$, i.e.,

$$Steps^{\mathcal{I}}(s) = \{\mu \mid \sum_{t \in S^{\mathcal{I}}} \mu(t) = 1 \text{ and } \underline{\mathbf{P}}(s,t) \leq \mu(t) \leq \overline{\mathbf{P}}(s,t) \text{ for all } t \in S^{\mathcal{I}}\}$$

for all $s \in S^{\mathcal{I}}$. In other words, under the *IMDP* semantic, any probability function from $Steps^{\mathcal{I}}(s)$ can be chosen before each transition.

TODO:
vllt zu stark eine Doppelung des vorherigen Satzes?; & ist "entered" ein passendes wort?

Furthermore, $Steps^{\mathcal{I}} = \bigcup_{s \in S^{\mathcal{I}}} Steps^{\mathcal{I}}(s)$ and $\mu(s,t)$ denotes the probability that the probability distribution $\mu(s)$ assigns to t, for $s,t \in S^{\mathcal{I}}$. This choice of a probability function is made non-deterministically by an external environment.

In [Bad] this external environment is defined as the *nature*.

Definition 4.6 (Nature)
A *nature* for an IDTMC $\mathcal{I}$ is a function $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ that maps each finite path on to a valid transition probability function, i.e., $\eta(\pi) \in Steps^{\mathcal{I}}(last(\pi))$.

Intuitively, this implies that η, i.e., the external environment, depends on all states that the model has visited up until $last(\pi)$ for any given $\pi \in Paths^{\mathcal{I}}_{fin}$. Notice, that we only define this external environment for the transition probability function and not the initial distribution. Recall, that results for the *fixed-initial-state definition* can be applied to our definition. Thus, it suffices to define this external environment for the transition probability function. For the rest of this chapter, we fix a nature $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$.

By applying a given *nature* η to $\mathcal{I}$, we resolve the uncertainty of the probability function, as η specifies a probability function $\mathbf{P} \in Steps^{\mathcal{I}}(s)$ for every finite path in $\mathcal{I}$. Thus, η induces a DTMC as follows.

Definition 4.7 (DTMC of an IDTMC Induced by a Nature)
For an IDTMC $\mathcal{I}$ and a nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (Paths^{\mathcal{I}}_{fin}, \mathbf{P}^{\mathcal{D}_\eta}, s_{in}) \ ,$$

where for all $\pi \in Paths^{\mathcal{I}}_{fin}$, $s \in S^{\mathcal{I}}$, and $\eta(\pi) = \mu(last(\pi))$:

$$\mathbf{P}^{\mathcal{D}_\eta}(\pi, \pi') = \begin{cases} \mu(last(\pi), s) & \text{if } \pi' = (\pi[0], \pi[1], \dots, last(\pi), s) \\ 0 & \text{otherwise.} \end{cases}$$

This definition is based on [BK08].

Notice, that these induced DTMCs are infinite in size, i.e., $|\mathcal{D}_\eta| = \infty$ and that, intuitively, every state $s \in S^{\mathcal{D}_\eta}$ is visited only once, as the states represent paths. Thus, $\mathcal{D}_\eta$ is an acyclic DTMC.

To relate the states in the induced DTMCs $\mathcal{D}_\eta$ to the states in the IDTMC $\mathcal{I}$, we define the function

$$\tau : Paths^{\mathcal{I}}_{fin} \to S^{\mathcal{I}}, \ (\pi[0], \dots, last(\pi)) \mapsto last(\pi) \ .$$

In other words, the states in the induced DTMCs that correspond to paths in $\mathcal{I}$ ending in some state $s \in S^{\mathcal{I}}$, are mapped to s. Intuitively,

$$\tau^{-1}(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid \tau(\pi) = s\}$$

defines the set of all states in the induced DTMCs that correspond to paths in $\mathcal{I}$ ending in s for all $s \in S^{\mathcal{I}}$. Moreover, we denote $\tau^{-1}(T) = \bigcup_{t \in T} \tau^{-1}(t)$ for $T \subseteq S^{\mathcal{I}}$.

TODO:
vllt brauche ich dieses set nicht

TODO:
zeigen dass well-defined?; & auch eine neue sache von mir (contribution)

TODO:
vllt muss ich die funktionsdef. mehr hervorheben oder iwie referenzieren?

TODO:
statt Pathsfin einfach $S^{\mathcal{D}_\eta}$ oder nochmal anmerken zumindest

Similarly, we define the set

$$\rho(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid last(\pi) = s \wedge \nexists k \in \underline{|\pi| - 1} : \pi[k] = s\} \ .$$

In other words, this set contains all states in $\mathcal{D}_\eta$ that correspond to finite paths in $\mathcal{I}$ which end in s and contain s precisely once. We denote $\rho(T) = \bigcup_{t \in T} \rho(t)$ for $T \subseteq S^{\mathcal{I}}$.

Importantly, however, certain types of natures, see Def. 4.8, can be represented by finite-state DTMCs.

TODO:
vllt auch noch MC-aug zitieren

Of particular relevance to this thesis are *stationary natures*. The following definition is based on [Bad].

Definition 4.8 (Stationary Nature)
A nature η for an IDTMC $\mathcal{I}$ is *stationary* if for all paths $\pi, \pi' \in Paths^{\mathcal{I}}_{fin}$, it holds that

$$last(\pi) = last(\pi') \implies \eta(\pi) = \eta(\pi') \ .$$

Intuitively, this implies that η only depends on the current state of the model, i.e., the state the model is in. Thus, stationary natures are functions of the form

$$\eta : S^{\mathcal{I}} \to Steps^{\mathcal{I}} \ .$$

Notice that the induced DTMCs of these stationary natures are finite in size.

Definition 4.9 (DTMC of an IDTMC Induced by a Stationary Nature)
For an IDTMC $\mathcal{I}$ and a stationary nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_\eta}, s_{in}) \ ,$$

where for all $s_i, s_j \in S^{\mathcal{I}}$ and $\eta(s_i) = \mu(s_i)$:

$$\mathbf{P}^{\mathcal{D}_\eta}(s_i, s_j) = \mu(s_i, s_j)$$

TODO:
quelle finden; zudem definition müssste ohne case unterscheidung ok sein, weil η ja passend definiert

Crucially, a stationary nature can thus be represented by a DTMC of the form Def. 4.9 as well as Def. 4.7. However, as all natures can be represented by the DTMCs of the form Def. 4.7, we base all defintions for IDTMCs under the IMDP semantic in this thesis on Def. 4.7.

Using these notions, we can now define the *IMDP semantic*.

Definition 4.10 (IMDP Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ under the IMDP semantic if and only if it is induced by a *nature* $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ on $\mathcal{I}$.

TODO:
irgendwas in richtung: Notice, that not only DTMCs of form ... induced by nature, also ... (stationary nature referenzieren); klar machen das die infinite definition nicht alle satisfying DTMCs darstellt

If $\mathcal{D}$ satisfies $\mathcal{I}$ under the IMDP semantic, we denote this as $\mathcal{D} \models_{\mathrm{I}} \mathcal{I}$.

The other common definition can be found in [BDL+17] and [BDLM21].

TODO:
maybe remove this

TODO:
Besseres example; vllt mittels finite memory scheduler; oder infinite induzierten DTMC von Katoen klauen

Example 4.11
Consider the IDTMC $\mathcal{I}$ in Fig. 4.1. The DTMC $\mathcal{D}_I$ depicted in Fig. 4.3 satisfies $\mathcal{I}$ under the IMDP semantic, i.e., $\mathcal{D}_I \models_I \mathcal{I}$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution

$$\iota_{init} = (0.6, 0, 0, 0, 0)$$

TODO:
Gleiche reach. prob. für diese Konstrukion und zugehöriger IDTMC in paper gezeigt (denke ich), weil reach prob $\subset$ PCTL. Und die Konstruktion ist für PCTL?

TODO:
2. Beispiel für : IMDP erlaubt mehr "behaviour" (Beispiel auf GoodNotes)

TODO:
besserer Übergang?

If we want to emphasize that a DTMC $\mathcal{D}$ satisfies the IDTMC $\mathcal{I}$ under both semantics, we write $\mathcal{D} \models \mathcal{I}$. Furthermore, if two DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ satisfy a given IDTMC $\mathcal{I}$, we write $\mathcal{D}_1, \mathcal{D}_2 \models \mathcal{I}$ for short.

Lastly, we introduce some notions of IDTMCs, which we use in the following chapters.

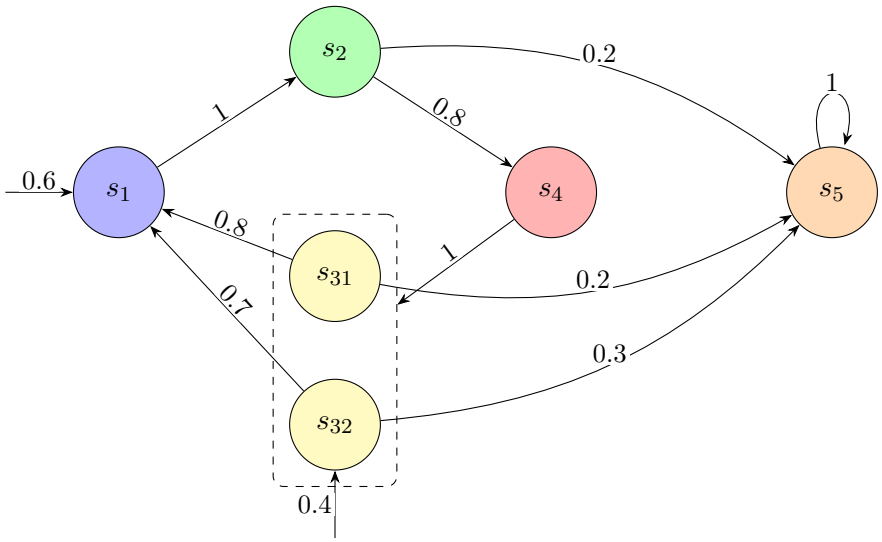


Figure 4.3: DTMC $\mathcal{D}_I$ with $\mathcal{D}_I \models_I \mathcal{I}$.

TODO:
cref the chapter

TODO:
Erst Probability Space einführen; Brauche das glaube ich nicht

First, we introduce reachability probabilities for IDTMCs. We base our definition of the quantitative reachability property in IDTMCs on [BDL+17].

Definition 4.12 (Reachability Probability in IDTMCs)
Given an IDTMC $\mathcal{I}$, a set of states $T \subseteq S^{\mathcal{I}}$, $p \in [0,1]$, and $\sim \in \{\leq, \geq\}$, then a quantitative reachability property on $\mathcal{I}$ under the UMC semantic is defined as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, if and only if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond T) \sim p$. Further, we define the quantitative reachability property on $\mathcal{I}$ under the IMDP semantic as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, if and only if there exists at least one DTMC $\mathcal{D} \models_I \mathcal{I}$ such that $Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t) \sim p$ under the IMDP semantic.

TODO:
check whether definition for IMDP is good

Additionally, we use the notation $Pr^{\mathcal{I}}_s(\Diamond T)$ to denote the reachability probability, when fixing $s \in S^{\mathcal{I}}$ as the unique initial state.

In this thesis, we are particularly interested in the optimal reachability probabilities, which we define as follows.

Definition 4.13 (Maxmimum and Minimum Reachability Probability) Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, we define the maximum reachability as $\sup_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ and the minimum reachability as $\inf_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ under the UMC semantic.

Furthermore, we define the maximum reachability as $\sup_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ and the minimum reachability as $\inf_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ under the IMDP semantic.

To shorten the notation, we denote the maximum reachability as $\overline{Pr}^{\mathcal{D}}(\Diamond s)$ and the minimum reachability as $\underline{Pr}^{\mathcal{D}}(\Diamond s)$. Moreover, we use the notation $\overline{Pr}^{\mathcal{I}}_s(\Diamond T)$ and $\underline{Pr}^{\mathcal{I}}_s(\Diamond T)$ to denote the maximum and minimum reachability probabilities, when fixing $s \in S^{\mathcal{I}}$ as the unique initial state.

Furthermore, we define transient and recurrent states. As we observed for the infinite-state DTMCs $\mathcal{D}_\eta$ defined in Def. 4.7, every state $s \in S^{\mathcal{D}_\eta}$ is visited only once. Thus, intuitively $Pr_s(\bigcirc\Diamond s) = 0$ for all $s \in S^{\mathcal{D}_\eta}$, meaning $S^{\mathcal{D}_\eta}$ contains no recurrent states. Consequently, we define transient and recurrent states exclusively for the UMC semantic.

Definition 4.14 (Transient and Recurrent states in IDTMCs) Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$ is called recurrent if and only if $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) = 1$.

Furthermore, a state $s \in S^{\mathcal{I}}$ is called transient if and only if $\underline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) < 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) < 1$.

We denote the set of recurrent states as $S^{\mathcal{I}}_{re}$ and the set of transient states as $S^{\mathcal{I}}_{tr}$.

Notice, that in contrast to DTMCs, there exist IDTMCs with $S^{\mathcal{I}}_{tr} \cap S^{\mathcal{I}}_{re} \neq \emptyset$. To observe this, consider state s_4 of Example 4.2. Let $\mathcal{D}_{tr} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{tr}}(s_4, s_3) = 1$. Therefore, $s_4 \in S^{\mathcal{D}_{tr}}_{tr}$ and consequently $s_4 \in S^{\mathcal{I}_E}_{tr}$. Now, let $\mathcal{D}_{re} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{re}}(s_4, s_4) = 1$. Then, $s_4 \in S^{D_{re}}_{re}$ and thus $s_4 \in S^{\mathcal{I}_E}_{re}$. Thus, $s_4 \in S^{\mathcal{I}_E}_{re} \cap S^{\mathcal{I}_E}_{tr}$.

TODO:
vllt abbildung/Beispiel duplizieren und hier nochmal zeigen

Due to this, we additionally define the sets

$$S_{\forall re} = \{s \in S^{\mathcal{I}} \mid \forall \mathcal{D} \models_U \mathcal{I} : Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) = 1 \wedge Pr^{\mathcal{D}}(\Diamond s) > 0\}$$

and

$$S_{\forall tr} = \{s \in S^{\mathcal{I}} \mid \forall \mathcal{D} \models_U \mathcal{I} : Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) < 1 \wedge Pr^{\mathcal{D}}(\Diamond s) > 0\} \ .$$

Intuitively, these are the sets that contain all states that are reachable and either recurrent or transient, respectively, in all DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic.

4.1 Relation between the semantics

In this section, we show some relations between the UMC and IMDP semantic that are relevant for the rest of this thesis.

Intuitively, the *IMDP* semantic is more expressive than the *UMC* semantic. More specifically for any given IDTMC $\mathcal{I}$, a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ according to the UMC semantic, i.e., $\mathcal{D} \models_U \mathcal{I}$, does also satisfy $\mathcal{I}$ according to the IMDP semantic, i.e., $\mathcal{D} \models_I \mathcal{I}$.

Theorem 4.15
For any IDTMC $\mathcal{I}$ and DTMC $\mathcal{D}$, if $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$.

TODO:
maybe ref Quant-analysis here; however they of course use different definition of IMDP semantic

We shortly prove this theorem.

Proof 4
Given an IDTMC $\mathcal{I}$ and a DTMC $\mathcal{D} \models_U \mathcal{I}$, there exists at least one stationary nature η_{st} and corresponding induced DTMC $\mathcal{D}_{\eta_{st}}$ of the form Def. 4.9, such that $\mathcal{D} = \mathcal{D}_{\eta_{st}}$.

In other words the set of induced DTMCs by stationary natures on $\mathcal{I}$ of the form Def. 4.9 is identical to the set of DTMCs that satisfy $\mathcal{I}$ under the UMC semantic.

To show this, we first define $\eta_{st}(s) \coloneqq \mathbf{P}^{\mathcal{D}}(s)$ for all $s \in S^{\mathcal{I}} = S^{\mathcal{D}}$, where $\mathbf{P}^{\mathcal{D}}(s_i)$ denotes the probability distribution of $\mathbf{P}^{\mathcal{D}}$ on s_i. Thus,

$$\mathbf{P}^{\mathcal{D}_{\eta_{st}}}(s_i, s_j) = \eta_{st}(s_i, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$$

for all $s_i, s_j \in S^{\mathcal{I}}$, where $\mu(s_i) = \eta(s_i)$. In other words, $\mathcal{D}_{\eta_{st}}$ and $\mathcal{D}$ have the identical probability distribution function. Moreover, $S^{\mathcal{D}_{\eta_{st}}} = S^{\mathcal{I}} = S^{\mathcal{D}}$ by definition, see Def. 4.9 and Def. 4.4. It suffices to show that the transition probability functions are identical, as we can apply results for the *fixed-intial-state-definition* to our own definition. Hence, $\mathcal{D}_{\eta_{st}} = \mathcal{D}$ and as $\mathcal{D}_{\eta_{st}} \models_I \mathcal{I}$, see Def. 4.10, we can conclude that $\mathcal{D} \models_I \mathcal{I}$. □

TODO:
Kurz zusammenfassen, was die relation der zwei Semantiken für ne folge hat

TODO:
denote in beginning of chapter or section that this (relation) is own contribution

Another important relation between the two semantics, that we use in this thesis, is that the semantics coincide on the maximum and minimum reachability probability, see [BDL+17, CHK13].

TODO:
vllt nocht bisschen mehr text hierzu, weiß aber nicht was

Chapter 5

EVTs in IDTMCs

We continue in this chapter by defining *expected visiting times* (EVTs) for IDTMCs. As we discussed in the previous section, IDTMCs can be seen as a set of DTMCs that satisfy the IDTMC according to the given semantic. These different DTMCs can have varying EVTs for a given state s in the IDTMC, thus we are interested in the upper and lower bound for the EVTs of s, i.e., in the largest and smallest EVT of s among the satisfying DTMCs. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$, a nature $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$, and stationary nature $\eta_{st} : S\mathcal{I} \to Steps^{\mathcal{I}}$ for the rest of this chapter.

TODO:
mehr text darüber, was in diesem chapter ist

Definition 5.1 (EVTs of IDTMCs)
For a IDTMC $\mathcal{I}$, we define the *upper expected visiting time* (uEVT) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and *lower expected visiting time* (ℓEVT) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ for all states $s \in S^{\mathcal{I}}$, as follows for the UMC semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

and the IMDP semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$,

where $T = \tau^{-1}(s)$.

TODO:
zeigen, dass maximum/minim; wenn noch Zeit; müsste davon folgen, dass sich ja u/lEVT durch Bruch etc. halt über quant. reach berechnen lassen oder?

TODO:
nochmal kritisch testen, ob das so stimmt jetzt

5.1 Calculating Expected Visiting Times of IDTMCs

In this section, we define loose bound for the ℓEVTs and uEVTs and provide precise characterizations for specific ℓEVTs and uEVTs under the UMC semantic.

We start by defining the loose bounds. Notice, that the most intuitive upper and lower bounds for the uEVTs and ℓEVTs of IDTMCs under the UMC semantic can be derived by using Eq. (3.2) and the upper and lower transition probability functions, as well as the initial distributions defined in Def. 4.1.

Theorem 5.2 (Upper and Lower Bounds for uEVTs and lEVTs)
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, the uEVT of s is bounded from above by

$$\overline{\mathbb{E}}[vt_s] \leq \mathbb{E}_{upp}[vt_s] := \overline{\iota}_{init}(s) + \sum_{t \in S} \overline{\mathbf{P}}(t,s) \cdot \mathbb{E}_{upp}[vt_t]$$

and the ℓEVT of s is bounded from below by

$$\underline{\mathbb{E}}[vt_s] \geq \mathbb{E}_{low}[vt_s] = \underline{\iota}_{init}(s) + \sum_{t \in S} \underline{\mathbf{P}}(t,s) \cdot \mathbb{E}_{low}[vt_t]$$

for all $s \in S^{\mathcal{I}}$.

TODO:
kann ich einf s in S machen (kein tr oder re), weil für die, die immer re sind einfach unendlich ist? Alternativ: Case unterscheidung (z.b. wenn es re gibt, dann ist upp infty etc; und wahrscheinlich für semantiken noch unterscheiden)

Proof 5
For any IDTMC $\mathcal{I}$ there exists a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ under the UMC semantic, i.e., $\mathcal{D} \models_U \mathcal{I}$, and $\mathbb{E}^{\mathcal{D}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. As $\mathcal{D} \models_U \mathcal{I}$, it follows that $\mathbf{P}(s,s') \leq \overline{\mathbf{P}}(s,s')$ for all $s, s' \in S$.

Thus, $\mathbb{E}^{\mathcal{D}}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] \leq \overline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}_{upp}[vt_t] = \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$.

The identical argument can be made to prove the lower bound for the ℓEVT. □

TODO:
hier weiter machen; grapheigenschaften

TODO:
Vllt case unterscheidung notwendig, wegen infty der EVTs?

However, these bounds are not tight, as can be seen in the following example.

Example 5.3
For this take a look at the upper bound of the IDTMC provided in Fig. 5.1. Using Theorem 5.2, we can calculate that $\mathbb{E}_{upp}[vt_{s_4}] = 2$, as $\mathbb{E}_{upp}[vt_{s_1}] = \mathbb{E}_{upp}[vt_{s_2}] = \mathbb{E}_{upp}[vt_{s_3}] = 1$. However, as the IDTMC is acyclic, see Def. 3.10, every state can be visited at most once. Thus, the upper bound $\mathbb{E}_{upp}[vt_{s_4}]$ is clearly not tight. Similarly, $\mathbb{E}_{low}[vt_{s_4}] = 0$, however s_4 needs to be visited at least once. Thus, this lower bound $\mathbb{E}_{low}[vt_{s_4}]$ is not tight either.

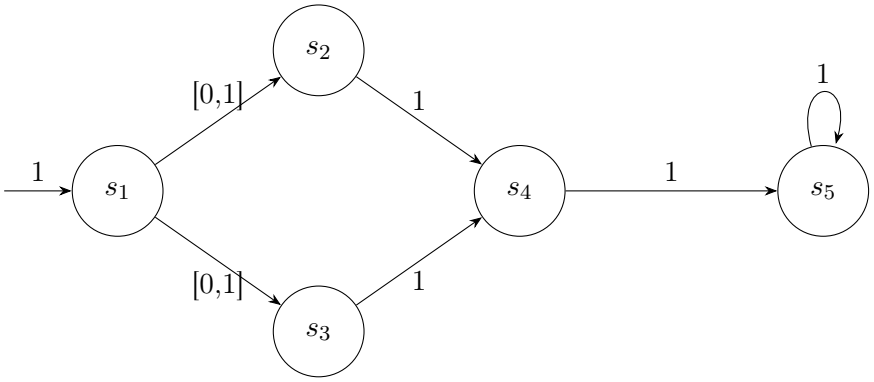


Figure 5.1: An example of a DTMC.

In the following, we therefore take a look at how we can characterize the precise values of the uEVTs and ℓEVTs.

First, we provide these characterizations under the UMC semantic, as our definition of transient and recurrent states in IDTMCs is limited to the UMC semantic, see Def. 4.14. Later, in Sec. 5.2, we extend these characterizations to the IMDP semantic.

TODO:
muss dieses extend dann noch schreiben in der section

Theorem 5.4
For a given state s in an IDTMC $\mathcal{I}$, $\overline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{re}$. Moreover, $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{\forall re}$.

Proof 6

TODO:
"We start by showing $\leftarrow$"

By our Assumption 1, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that s is reachable in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$. Moreover, as $s \in S^{\mathcal{I}}_{re}$, there exists a DTMC with $\mathcal{D}_{re} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{re}}_s(\bigcirc\Diamond s) = 1$, see Def. 4.14, thus $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Using Theorem 5.11, there exists a DTMC $\mathcal{D}_{+re} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{+re}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}_{+re}}_s(\bigcirc\Diamond s) = 1$, as $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Therefore, clearly $s \in S^{\mathcal{D}_{+re}}_{re}$, see Def. 3.11 and using Lemma 3.18, we can thus conclude that $\mathbb{E}^{\mathcal{D}_{+re}}[vt_s] = \infty$ and thus $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$.

Conversely, if $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$, then there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$, i.e., $s \in S^{\mathcal{D}}_{re}$, see Lemma 3.18. Thus, it follows directly that $s \in S^{\mathcal{I}}_{re}$, see Def. 4.14.

Moreover, $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{\forall re}$ as $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) > 0$ for all $\mathcal{D} \models_U \mathcal{I}$, by the definition of $S_{\forall re}$. Thus, using Lemma 3.18, we can directly conclude the proof. □

TODO:
benutze hier schon quant reach konstruktion. Brauche aber dieses theorem in quant reach section

Moreover, we can give a characterization for specific ℓEVTs.

Theorem 5.5
For any $s \in S^{\mathcal{I}}$ in a given IDTMC $\mathcal{I}$, $\underline{\mathbb{E}}[vt_s] = 0$ if and only if $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, i.e., there exists at least one satisfying DTMC of $\mathcal{D} \models_U \mathcal{I}$, such that s is not reachable in $\mathcal{D}$.

TODO:
erweitern um Expupper = 0 für s in Satr; wenn ich das hier mache, dann theorem als zwei claims gestalten und sagen claim 1 für UMC und IMDP und claim 2 nur für UMC

Proof 7

TODO:
"We start by showing ←"

If $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, then there exists $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond s) = 0$, i.e., there exists no path in $\mathcal{D}$ that contains s. As, therefore, $vt_s(\pi) = 0$ for all $\pi \in Paths^{\mathcal{D}}_{inf}$, we can conclude, using Def. 3.17, that $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$.

Conversely, if $\underline{Pr}^{\mathcal{I}}(\Diamond s) > 0$, then there exists $\pi \in Paths^{\mathcal{D}}_{inf}$ with $vt_s(\pi) > 0$ for all $\mathcal{D} \models_U \mathcal{I}$. Therefore, $\mathbb{E}^{\mathcal{D}}[vt_s] > 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] > 0$. □

TODO:
mittels case unterscheidung beide semantiken möglich; so nicht, weil muss zwischen s und tau(s) unterscheiden

TODO:
Vielleicht noch mehr dazu, wie sich bestimmen lässt, dass $s \in S^{\mathcal{I}}_{re}$ und s not reachable, mittels verschiedener Grapheigenschaften?

Remark. *Non-attainability of values in EVT intervals under the UMC semantic.*
In general, not every value in $[\underline{\mathbb{E}}[vt_s], \overline{\mathbb{E}}[vt_s]]$ is attainable under the UMC semantic. In other words, there exists an IDTMC $\mathcal{I}$ with a state $s \in S^{\mathcal{I}}$, such that for some $x \in [\underline{\mathbb{E}}^{\mathcal{I}}[vt_s], \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]]$ it holds that $\mathbb{E}^{\mathcal{D}}[vt_s] \neq x$ for all $\mathcal{D} \models_U \mathcal{I}$. This holds even for $\overline{\mathbb{E}}[vt_s] < \infty$.

We provide a short example for this remark.

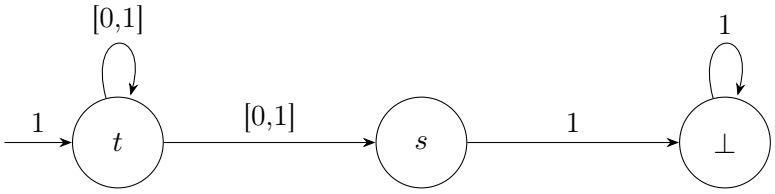


Figure 5.2: An example of a DTMC.

Example 5.6
Consider the IDTMC $\mathcal{I} = (\{t, s, \bot\}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \iota_{init}, \bar{\iota}_{init})$ shown in Fig. 5.2. Intuitively, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 1$. However, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = x$ for any $x \in (0, 1)$, as $\mathbb{E}^{\mathcal{D}}[vt_s] = 1$ if $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ and $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ else.

TODO:
noch weiter ausführen: Unterschied im Vergleich zu IMP Semantik erklären

5.2 Relating Expected Visiting Times and Quantitative Reachability

Similar to Sec. 3.3, we relate *expected visiting times* to *quantitative reachability* in this section. Moreover, we show that for any IDTMC $\mathcal{I}$, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ that has the optimal, i.e., maximum or minimum, reachability probability for all states in $\mathcal{I}$.

TODO:
muss das noch für IMDP folgern; folgt direkt weil ja -U auch -I.

As we concluded in Sec. 3.3, we can calculate the EVT for all transient states $s \in S^{\mathcal{D}}_{tr}$ in a DTMC $\mathcal{D}$ using quantitative reachability, see Lemma 3.21: $\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}}(\Diamond s)}{1-Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)}$. As defined in Def. 5.1, the uEVT and ℓEVT of $\mathcal{I}$, under the UMC semantic, are equal to the maximum and minimum EVT amongst the satisfying DTMCs $\mathcal{D} \models_U \mathcal{I}$. Thus, the uEVT and ℓEVT of $\mathcal{I}$ are given by the largest and smallest ratio of these quantitative reachability probabilites among the satisfying Markov Chains, as those optimal ratios define the maximum and minimum EVT.

TODO:
ich denke das kein noch etwas mehr ausgeführt werden

We continue by proving that for all states $s \in S^{\mathcal{I}}$ there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathcal{D}$ has maximal reachability probability for all states $t \in S^{\mathcal{I}}$ to s, i.e., $Pr^{\mathcal{D}}_t(s) = \overline{Pr}^{\mathcal{I}}_t(s)$. Conversely, we show the same for the minimum reachability probability.

Remark. *Existence of DTMC with optimal reachability.* For any finite IDTMC $\mathcal{I}$, there exists an optimal stationary nature η_{st} on $\mathcal{I}$ which induces a DTMC $\mathcal{D}_{\eta_{st}}$, such that $Pr^{\mathcal{D}_{\eta_{st}}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$ for all $t, s \in S^{\mathcal{I}}$, see [BK08, Lemma 10.102] and [Bad]. As $\mathcal{D}_{\eta_{st}} \models \mathcal{I}$, see Theorem 4.15, there thus exists a DTMC $\mathcal{D} \models \mathcal{I}$ with $Pr^{\mathcal{D}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$ for every $t, s \in S^{\mathcal{I}}$ which is precisely $\mathcal{D} = \mathcal{D}_{\eta_{st}}$. The same holds for the minimum reachability probability, see [BK08, Lemma 10.113].

TODO:
Soll das mit rein?: This follows directly from the compactness and continuity of the polytope of DTMCs satisfying $\mathcal{I}$; it is not reproved here, only reused.

Intermediate States. Using the notion of finite paths, we introduce *intermediate states.*

Definition 5.7 (Intermediate States)
Given a DTMC $\mathcal{D}$, a set of states $T \subseteq S$, and a state $s \in S$, then $u \in S$ is *intermediate* if and only if there exists path $\pi \in Paths^{\mathcal{D}}_{fin}$, such that $\pi[0] \in T$, $last(\pi) = s$, and $\pi[k] = u$ for some $k \in \underline{|\pi|}$.

In other words, there exists a finite path from some $t \in T$ to s in $\mathcal{D}$ that visits u. We denote the set of these intermediate states as $I^{\mathcal{D}}_{T,s}$ for a set of states $T \in S$ and state $s \in S$ in $\mathcal{D}$. Moreover, if T is a singleton set, i.e., $T = \{t\}$, then we simply denote the set of intermediate states as $I^{\mathcal{D}}_{t,s}$.

Theorem 5.8
Given an IDTMC $\mathcal{I}$, states $t, s \in S^{\mathcal{I}}$, and a DTMC $\mathcal{D}$ with $\mathcal{D} \models_U \mathcal{I}$.

If $\mathcal{D}$ has the maximum reachability probability from t to s, i.e., $Pr^{\mathcal{D}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$, then for all $u \in I^{\mathcal{D}}_{t,s}$:

$$Pr^{\mathcal{D}}_u(\Diamond s) = \overline{Pr}^{\mathcal{I}}_u(\Diamond s)$$

Conversely, if $\mathcal{D}$ has the minimum reachability probability from t to s, i.e., $Pr^{\mathcal{D}}_t(\Diamond s) = \underline{Pr}^{\mathcal{I}}_t(\Diamond s)$, then for all $u \in I^{\mathcal{D}}_{t,s}$:

$$Pr^{\mathcal{D}}_u(\Diamond s) = \underline{Pr}^{\mathcal{I}}_u(\Diamond s)$$

In other words, if the reachability probability, under the UMC semantic, from t to s is optimal, then the reachability probability of all intermediate states is optimal as well.

TODO:
Hannah fragen was nochmal problem für IMDP Semantik ist und gegebenenfalls dann den Beweis erweitern

Proof 8
As stated in the previous remark, there exists at least one DTMC $\mathcal{D}_\eta \models_U \mathcal{I}$ with $Pr^{\mathcal{D}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$ for every $t, s \in S^{\mathcal{I}}$, such that $\mathcal{D}_\eta$ is induced by a stationary nature η on $\mathcal{I}$.

Based on [BK08, Theorem 10.100], we can formulate the following equation which characterizes the maximum reachability probability of IDTMCs under the IMDP semantic.

$$\overline{Pr}^{\mathcal{I}}_t(\Diamond s) = \max_{\mathbf{P} \in Steps^{\mathcal{I}}(t)} \left\{ \sum_{u \in N(t)} \mathbf{P}(t,u) \cdot \overline{Pr}^{\mathcal{I}}_u(\Diamond s) \right\} . \tag{5.1}$$

TODO:
geht diese formel so immer noch, weil Steps ja unendlich groß (uncountable?)

Intuitively, the DTMCs $\mathcal{D}_{\eta'}$ that are induced by natures η' which solve this equation have the maximum reachability probability $Pr_u^{\mathcal{D}_{eta}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t,s}^{\mathcal{D}}$, as shown in [BK08]. As $\mathcal{D}_\eta$ has the maximum reachability probability of t, when starting in s, and $\mathcal{D}_\eta \models_I$, see theorem 4.15, $\mathcal{D}_\eta$ solves Eq. (5.1).

Conversely, the same can be reasoned for minimum reachability using [BK08, Theorem 10.109]. □

Corollary 5.9
Given an IDTMC $\mathcal{I}$, two transient states $t_1, t_2 \in S^{\mathcal{I}}$, a (goal) state $s \in S^{\mathcal{I}}$, and two DTMCs $\mathcal{D}_1, \mathcal{D}_2$ with $\mathcal{D}_1, \mathcal{D}_2 \models_U \mathcal{I}$ and $Pr_{t_1}^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$ and $Pr_{t_2}^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_{t_2}^{\mathcal{I}}(\Diamond s)$. In other words the DTMCs have a maximum reachability probability for t_1 and t_2, respectively.

For all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$, the reachability probabilities $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ are identical.

Conversely, the same is true for minimum reachability probabilities.

TODO:
vllt länger ausführen

In other words, if $\mathcal{D}_1$ and $\mathcal{D}_2$ have optimal reachability from t_1 and t_2, respectively, to s, then all states $u \in S^{\mathcal{I}}$ that are intermediate states of t_1 and s as well as t_2 and s have the identical reachability probability in $\mathcal{D}_1$ and $\mathcal{D}_2$.

Proof 9
This corollary follows from Theorem 5.8. Using Theorem 5.8, we can conclude that $Pr_u^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1}$ and $Pr_u^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_2,s}^{\mathcal{D}_2}$. Thus, $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$.

Conversely, this holds for minimum reachability probabilities using the same reasoning.

TODO:
gut/ausreichend so?

Lemma 5.10 (One-state Extension)
Given a set $T \subset S$, a state $t \notin T$, and a state s, assume that $\mathcal{D}_T \models_U \mathcal{I}$ with

$Pr_u^{\mathcal{D}_T}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $\mathcal{D}_t \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. Define $\mathcal{D}_{cmb} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{cmb}}, \iota_{init}^{\mathcal{D}_T})$ by

$$\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \begin{cases} \mathbf{P}^{\mathcal{D}_t}(w,v) & \text{if } w \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T} \\ \mathbf{P}^{\mathcal{D}_T}(w,v) & \text{else.} \end{cases}$$

Stated in words, $\mathcal{D}_{cmb}$ is constructed such that its probability distribution function is equal to that of $\mathcal{D}_t$ in all states that are only intermediate states for t and not T, while it is equal to that of $\mathcal{D}_T$ in all other states.

Then, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$, i.e., $\mathcal{D}_{cmb}$ has the maximum reachability probability for all states in $T \cup \{t\}$.

Conversely, if $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$, the identical construction of $\mathcal{D}_{cmb}$ implies $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

TODO:
"implies"?

TODO:
notation: warum Pr für alle u in T, wenn auch einfach Pr-T machen kann?

Intuitively, this lemma states that using two DTMCs $\mathcal{D}_T$ and $\mathcal{D}_t$ which have optimal reachability probabilities from T and t to s, respectively, we can construct a DTMC $\mathcal{D}_{cmb}$ that has optimal reachability from $T \cup \{t\}$ to s.

Proof 10
First, as $\mathcal{D}_t, \mathcal{D}_T \models_U \mathcal{I}$ it follows from the construction that also $\mathcal{D}_{cmb} \models \mathcal{I}$. Next, as for each $w \in S^{\mathcal{I}}$ either $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_t}(w,v)$ or $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_T}(w,v)$ for all $v \in S^{\mathcal{I}}$ it follows that $\mathbf{P}^{\mathcal{D}_{cmb}}$ is well-defined, i.e., a discrete probability distribution, see Def. 2.8.

As $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_T}(w,v)$ for all $w \in I_{T,s}^{\mathcal{D}_T}$, $\overline{Pr}_u^{\mathcal{I}}(\Diamond s) = Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T$, see Lemma 3.7.

Thus, we need to show that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. As we know that $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, we will prove that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_t^{\mathcal{D}_t}(\Diamond s)$. Notice, that if $I_{t,s}^{\mathcal{D}_t} \cap I_{T,s}^{\mathcal{D}_T} = \emptyset$, then the proof concludes using the same argument as for $u \in T$. Furthermore, if $I_{t,s}^{\mathcal{D}_t} \subseteq I_{T,s}^{\mathcal{D}_T}$, then the proof concludes directly using Theorem 5.8.

Recall, that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \sum_{u \in N(t) \cap I_{t,s}^{\mathcal{D}_{cmb}}} \mathbf{P}^{\mathcal{D}_{cmb}}(t,u) \cdot Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$, see Lemma 3.7.
For all $u \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T}$, $\mathbf{P}^{\mathcal{D}_{cmb}}(u,w) = \mathbf{P}^{\mathcal{D}_t}(u,w)$ and for all $u \in I_{T,s}^{\mathcal{D}_T} \cap I_{t,s}^{\mathcal{D}_t}$, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$, see Corollary 5.9. From this, we can conclude that $Pr_t^{\mathcal{D}_t}(\Diamond s) = Pr_t^{\mathcal{D}_{cmb}}(\Diamond s)$. Thus, we have shown that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Conversely, the same holds for minimum reachability probabilities using this reasoning, given that $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$.□

Crucially, Lemma 5.10 can be extended to all states.

TODO:
wirklich?

Theorem 5.11
Given an IDTMC $\mathcal{I}$ and a state $s \in S^{\mathcal{I}}$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all states $t \in S^{\mathcal{I}}$.

TODO:
extend theorem to minimum reach prob

Proof 11
Consider any given enumeration $t_1, t_2, \ldots$ of $S^{\mathcal{I}}$. Clearly, this enumeration is finite, since $S^{\mathcal{I}}$ is finite, see Assumption 1.

Construction. Now, let $\mathcal{D}^{(1)} \models_U \mathcal{I}$ with $Pr_{t_1}^{\mathcal{D}^{(1)}}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}$ and $T_1 := I_{t_1,s}^{\mathcal{D}^{(1)}}$, trivially such a DTMC must exist. By Theorem 5.8, $Pr_u(\Diamond s) = \overline{\mathbf{P}}_u(\Diamond s)$.

Inductively, $\mathcal{D}^{(n)} \models_U \mathcal{I}$ (and $s \in T_n$). If $t_{n+1} \in T_n$, then we set $\mathcal{D}^{(n+1)} := \mathcal{D}^{(n)}$ and $T_{n+1} = T_n$. If not, we use Lemma 5.10 with $T = T_n$, $\mathcal{D}_T = \mathcal{D}^{(n)}$, and $t = t_{n+1}$ to obtain $\mathcal{D}^{(n+1)} := \mathcal{D}_{cmb}$ with $\mathcal{D}^{(n+1)} \models_U \mathcal{I}$. We set $T_{n+1} = T_n \cup I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$, thus and $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_{n+1}$, as $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_n$ (induction), $Pr_{t_{n+1}}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_{n+1}}^{\mathcal{I}}(\Diamond s)$ (see Lemma 5.10), and $Pr_u^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$ (see Theorem 5.8). □

Thus, to conclude, for any given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, there exist $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ with maximum and minimum reachability probability, respectively, for all states $t \in S^{\mathcal{I}}$.

TODO:
vllt am Anfang dieser Beweiskette ein theorem, dass genau das hier statet

Recall, that the ℓEVTs and uEVTs under the UMC semantic are given by the optimal ratios of Lemma 3.21: $\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}}(\bigcirc \Diamond s)}$, as we concluded in the beginning of this section. Thus, we can conclude the following from the existence of $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$.

Theorem 5.12
Given an IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$:

(i) There exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$.

Conversely, given $\mathcal{D}_{min}$, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] > \mathbb{E}^{\mathcal{D}}[vt_s]$.

In other words $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ under the UMC semantic.

(ii) The ℓEVT and uEVT of s are both identical under the two semantics, i.e., the semantic does not have any impact on the value of the ℓEVT and uEVT.

Proof 12
We start by proving (i).

As we showed in Theorem 5.11, there exist DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ for any given IDTMC $\mathcal{I}$ that have maximum or minimum reachability from all states to s, i.e.,

$$Pr_t^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s) \qquad \text{or} \qquad Pr_t^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$$

for all $t \in S^{\mathcal{I}}$. We fix these two DTMCs for rest of this proof. Further, we can thus conclude that

$$Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) \tag{5.2}$$

or

$$Pr^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ . \tag{5.3}$$

Using Lemma 3.21, we can thus differentiate the following three cases for s:

1. If $s \in S^{\mathcal{I}}_{\forall re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$ and thus Lemma 3.21 is undefined for $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. However, using Theorem 5.4 we can conclude that $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$. Using Eq. (5.2) and Eq. (5.3), we can deduce that s is reachable and recurrent in both $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. Thus, using Lemma 3.18, we can conclude that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$.
2. If $s \in S^{\mathcal{I}}_{re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$ and thus Lemma 3.21 is undefined for $\overline{\mathbb{E}}[vt_s]$. However, using Theorem 5.4 we can conclude that $\overline{\mathbb{E}}[vt_s] = \infty$. Analogously to case (1), using Eq. (5.2) and Lemma 3.18 it follows that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$. For the calculation of $\underline{\mathbb{E}}[vt_s]$, see case (3).

3. In all remaining cases, we can calculate the ℓEVT and uEVT using Lemma 3.21

$$\underline{\mathbb{E}}[vt_s] = \frac{\underline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \underline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{min}}(\Diamond s)}{1 - Pr^{\mathcal{D}_{min}}_s(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{min}}[vt_s]$$

and

$$\overline{\mathbb{E}}[vt_s] = \frac{\overline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{max}}(\Diamond s)}{1 - Pr^{\mathcal{D}_{max}}_s(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{max}}[vt_s] \ .$$

As $s \notin S^{\mathcal{I}}_{re}$, by definition, there exists no DTMC $\mathcal{D} \models_U \mathcal{I}$, such that $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) = 1$. In other words, $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) < 1$ for all $\mathcal{D} \models_U \mathcal{I}$. Thus, the equations are well-defined.

We continue by shortly proving the equations presented in case (3) by contradiction. Let us assume that there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$. Using Lemma 3.21, it thus follows that either

$$Pr^{\mathcal{D}}(\Diamond s) > Pr^{\mathcal{D}_{max}}(\Diamond s) \qquad \text{or} \qquad Pr^{\mathcal{D}}(\bigcirc\Diamond s) > Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s) \ .$$

However, as $Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ this leads to a contradiciton. The proof for $\mathcal{D}_{min}$ follows analogously by reversing the inequalities.

TODO:
folgendes evtl. noch weiter ausführen: Für induced DTMC (IMDP Semantik), ist reach prob = EVT, weil azyklisch; l/uEVT sind ja dann Summe der EVTs → DTMC reach-zu-EVT Formel funktioniert, weil return dann einfach immer 0 und die einzelnen reaches für l/uEVT zusammenaddiert werden; falls ich das hier noch machen, dann sollte das in die relating EVT und reach section

Lastly, we prove (ii). Recall, that for any DTMC $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$, see Theorem 4.15. Thus, $\mathcal{D}_{max}, \mathcal{D}_{min} \models_I \mathcal{I}$. Let us assume that η_{max} and η_{min} are the stationary natures that induce $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. Further, let us assume that $\mathcal{D}_{\eta_{max}}, \mathcal{D}_{\eta_{min}} \models \mathcal{I}$ are the acyclic infinite-state representations of η_{max} and η_{min}, see Def. 4.7.

As stated in [MKQW25], the EVTs of states $s \in S$ in acyclic DTMCs $\mathcal{D}$ coincide with the reachability probability of s, i.e., $\mathbb{E}^{\mathcal{D}}[vt_s] = Pr^{\mathcal{D}}(\Diamond s)$. Thus, the expected visiting times of the states $t \in S^{\mathcal{D}}_{\eta_{max}}$ and $t' \in S^{\mathcal{D}}_{\eta_{min}}$ for $\mathcal{D}_{\eta_{max}}$ and $\mathcal{D}_{\eta_{min}}$ are given by,

$$\mathbb{E}^{\mathcal{D}_{\eta_{max}}}[vt_t] = Pr^{\mathcal{D}_{\eta_{max}}}(\Diamond t) \qquad \text{and} \qquad \mathbb{E}^{\mathcal{D}_{\eta_{min}}}[vt'_t] = Pr^{\mathcal{D}_{\eta_{min}}}(\Diamond t') \ .$$

Now notice, that as $\mathcal{D}_{max}$ has the maximum reachability probability of s from all $t \in S^{\mathcal{I}}$ and the two semantics coincide on maximal reachability probabilities, see

Sec. 4.1, it intuitively follows that there exists no $\mathcal{D} \models_I \mathcal{I}$

$$\sum_{\pi \in \tau^{-1}(s)} Pr^{\mathcal{D}}(\Diamond \pi) > \sum_{\pi \in \tau^{-1}(s)} Pr^{\mathcal{D}_{max}}(\Diamond \pi) \ . \tag{5.4}$$

!!! We shortly prove this by contradiction. Let us assume that there exists a DTMC $\mathcal{D} \models_I \mathcal{I}$, such that Eq. (5.4) holds. Thus, there exists at least one $\pi \in \tau^{-1}(s)$, such that $Pr^{\mathcal{D}}(\Diamond \pi) > Pr^{\mathcal{D}_{max}}(\Diamond \pi)$ ending in s

TODO:
wenn möglich/vllt das noch besser/länger ausführen

Consequently,

$$\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}\Big(\bigcup_{t \in \tau^{-1}(s)} \Diamond t \Big) \qquad \text{and} \qquad \overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}\Big(\bigcup_{t \in \tau^{-1}(s)} \Diamond t \Big) \ ,$$

□

see Def. 5.1.

TODO:
das alles hier auf jeden fall nochmal tiefgründig überprüfen

5.2.1 UMC semantic

In this section we examine how the precise values of the uEVTs and lEVTs can be calculated for IDTMCs under the UMC semantic.

As defined, there can be multiple DTMCs $\mathcal{D}$ that satisfy an IDTMC $\mathcal{I}$ with differing transition probability functions $\mathbf{P}^{\mathcal{D}}$ and initial distributions $\iota^{\mathcal{D}}_{init}$. To determine the uEVT and ℓEVT for a given state $s \in S^{\mathcal{I}}$, we thus have to find the satisfying DTMCs that maximize and minimze Eq. (3.2), respectively. In other words, we need to solve the equations

$$\mathbb{E}[vt_s] = \mathrm{opt}_{\mathcal{D} \models \mathcal{I}} \left(\iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}^{\mathcal{D}}(t,s) \cdot \mathbb{E}[vt_t] \right), \qquad \mathrm{opt} \in \{\min, \max\}.$$

In [CHK13], the authors construct the following NLP (which we have rewritten slightly) to compute any maximum or minimum reachability, respectively, for a set of nodes $T \subseteq S$, using the fixed-initial-state-definition.

$$
\begin{aligned}
&\text{opt} && \sum_{s \in S \setminus T} x_s && \text{opt} \in \{\max, \min\} && \\
&\text{s.t.} && x_s \sim \sum_{s' \in S \setminus T} \mathbf{P}(s,s') x_{s'} + \sum_{s' \in T} \mathbf{P}(s,s') && \forall s \in S \setminus T \text{ and } \sim \in \{\leq, \geq\} && (1) \\
& && x_s = 1 && \forall s \in T && (2) \\
& && \sum_{s_j \in N(s)} \mathbf{P}(s_i, s_j) = 1 && \forall s_i \in S_{tr} && (3) \\
& && \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \mathbf{P}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) && \forall s_i, s_j \in S^{\mathcal{I}}_{tr} && (4)
\end{aligned}
$$

(Allowed only max + $\leq$ and min + $\geq$)

TODO:
alternatively can be calculated over LP or VI for max./min. reachability probabilities in MDPs for memoryless schedulers; gezeigt in der Katoen-Bibel

As shown in [CHK13], this NLP can be solved in PTIME. Thus given an IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, we can calculate the maximum or minimum reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc \Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc \Diamond s)$ of a state s in PTIME.

Recall, that the ℓEVT and uEVT of s can be calculated through a polynomial function (fraction) for the UMC semantic using only $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc \Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc \Diamond s)$, see Theorem 5.12.

Thus, we can calculate the ℓEVT and uEVT of s in PTIME for the UMC semantic.

TODO:
vllt das hier noch weiter ausführen & als theorem?

5.2.2 IMDP semantic

In this subsection we will look at how uEVTs and lEVTs can be calculated for the IMDP semantic.

By combining our conclusion that the ℓEVT and uEVT of a given state $s \in S^{\mathcal{I}}$ under the UMC semantic can be calculated in PTIME and claim (ii) of Theorem 5.12, we can intuitively conclude that the ℓEVT and uEVT of s under the IMDP semantic can be calculated in PTIME as well.

In the following, we present a different approach to prove this result. By showing this approach both the relation between the reachability probabilities and expected visiting times as well as the differences between the UMC and IMDP semantic becom more apparant. Moreover, this result allows us to infer some additional results for the IMDP semantic.

TODO:
based on this concept of forward/backward pass: can understand why acyclical is easy –> necessarily do not interfere; but useless info ig

First, we define the DTMCs

$$\mathcal{D}_{s_{in}\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}, \iota_{init}^{\mathcal{D}_{s_{in}\to s}}) \qquad \text{and} \qquad \mathcal{D}_{s\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s\to s}}, \iota_{init}^{\mathcal{D}_{s\to s}})$$

such that $\mathcal{D}_{s_{in}\to s}, \mathcal{D}_{s\to s} \models_U \mathcal{I}$ and

$$Pr^{\mathcal{D}_{s_{in}\to s}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s\to s}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ . \tag{5.5}$$

Recall, that a DTMC $\mathcal{D} \models_I \mathcal{I}$ satisfies an IDTMC $\mathcal{I}$ if and only if it is induced by a nature η_{aug} on $\mathcal{I}$. Notice, that we can define a nature that chooses the probability distribution specified by $\mathcal{D}_{s_{in}\to s}$ until state s is visited for the first time and then chooses the probability distribution specified by $\mathcal{D}_{s\to s}$. Formally, we can define this nature as follows:

$$\eta_{aug} : Paths_{fin}^{\mathcal{I}} \to Steps^{\mathcal{I}}$$

such that

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(last(\pi)) & \text{if } \nexists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s\to s}}(last(\pi)) & \text{otherwise.} \end{cases}$$

This nature can be characterized by having two stages where we transition from the first to the second stage when passing through s for the first time.

TODO:
ist underline(|pi|) richtig oder muss es |pi|-1 sein?

We continue by defining the DTMC $\mathcal{D}_{aug}^s$ using $\mathbf{P}^{\mathcal{D}_{s_{in}\to s}}$ and $\mathbf{P}^{\mathcal{D}_{s\to s}}$.

Construction. Given an IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, the previously defined $\mathcal{D}_{s_{in}\to s}$, $\mathcal{D}_{s\to s}$ (corresponding to either the suprema or infima) and a state s (for which we want to calculate the EVT), we define $\mathcal{D}_{aug}^s$, with $S^{\mathcal{D}_{aug}^s} = S_{st1} \cup S_{st2}$, where the state set for stage 1 is

$$S_{st1} = \{s^{(1)} \mid s \in S^{\mathcal{I}} = S^{\mathcal{D}_{s_{in}\to s}}\}$$

and the state set for stage 2 is

$$S_{st2} = \{s^{(2)} \mid s \in S^{\mathcal{I}} = S^{\mathcal{D}_{s\to s}}\}\ .$$

Furthermore,

$$\mathbf{P}^{\mathcal{D}_{aug}^s}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$$

for all $s_1^{(1)} \in S_{st1}\backslash\{s\}$, $s_2^{(1)} \in S_{st1}$ and $\mathbf{P}^{\mathcal{D}_{aug}^s}(s^{(1)}, s^{(2)}) = 1$. Thus, $\mathcal{D}_{aug}^s$ has the transition probability function of $\mathcal{D}_{s_{in}\to s}$ until s is reached for the first time and $\mathcal{D}_{aug}^s$

necessariliy transitions from the stage 1 copy of s, namely $s^{(1)}$, to the stage 2 copy of s, namely $s^{(2)}$.

Similarly,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(2)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_1, s_2)$$

for all $s_1^{(2)}, s_2^{(2)} \in S_{st2}$, i.e., $\mathcal{D}^s_{aug}$ has the transition probability function of $\mathcal{D}_{s\to s}$ after s has been reached for the first time. All other pairs of states are mapped to 0 by $\mathbf{P}^{\mathcal{D}^s_{aug}}$. Thus, $\mathcal{D}^s_{aug}$ cannot transition back to states of stage 1, after s has been entered.

Lastly,

$$\iota_{init}^{\mathcal{D}^s_{aug}}(t^{(1)}) = \iota_{init}^{\mathcal{D}_{s_{in}\to s}}(t)$$

for all $t^{(1)} \in S_{st1} = S^{\mathcal{D}_{s_{in}\to s}}$. All other states are mapped to 0 by $\iota_{init}^{\mathcal{D}^s_{aug}}$

TODO:
s-1, s-2 auf s-i,s-j ändern!

To illustrate this construction of $\mathcal{D}^s_{aug}$, we provide the following example.

Example 5.13
Recall the IDTMC $\mathcal{I}$ in Fig. 4.1, displayed again in Fig. 5.3. Fig. 5.4 shows $\mathcal{D}^{s_4}_{aug}$, i.e., the construction of the maximizing DTMC for the EVT of $\mathcal{I}$.

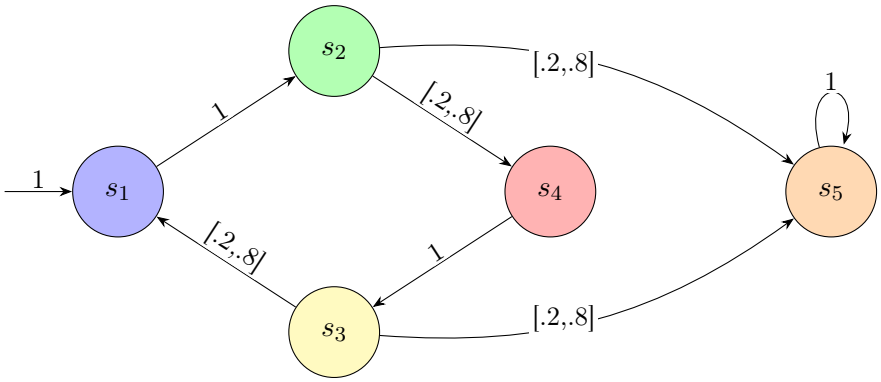


Figure 5.3: target state in red

TODO:
perhaps rewrite a bit after reformulation with mapping for quant. reach.

We now show that $\mathcal{D}^s_{aug}$ satisfies $\mathcal{I}$ under the IMDP semantic.

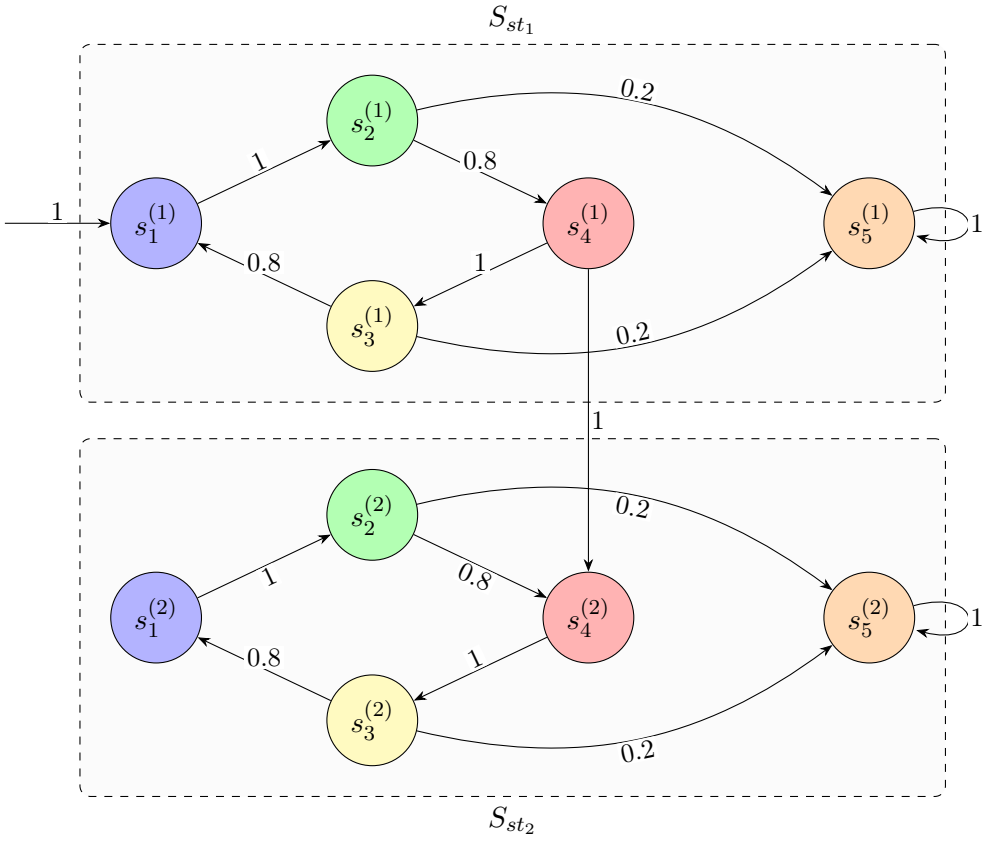


Figure 5.4: $\tau(t_1) = \tau(t_4) = s_1$, $\tau(t_2) = \tau(t_7) = s_2$, $\tau(t_5) = s_3$, $\tau(t_3) = \tau(t_6) = s_4$. Target state in red.

Lemma 5.14
Given and IDTMC $\mathcal{I}$, a state s, and the corresponding $\mathcal{D}^s_{aug}$, it holds that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$

Proof 13
We show that $\mathcal{D}^s_{aug}$ is induced by the nature η_{aug} on $\mathcal{I}$ defined by

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(last(\pi)) & \text{if } \nexists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s\to s}}(last(\pi)) & \text{otherwise.} \end{cases}$$

Since $\mathcal{D}_{s_{in}\to s} \models_U \mathcal{I}$ and $\mathcal{D}_{s\to s} \models_U \mathcal{I}$, both transition distributions are admissible in $\mathcal{I}$. Hence η_{aug} is a valid nature on $\mathcal{I}$.

Before the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s_{in}\to s}$. There-

fore, for all $s_i, s_j \in S$ with $s_i \neq s$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(1)}, s_j^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_i, s_j) = \eta_{\text{aug}}(\pi)(s_j)$$

for every finite path π ending in s_j that does not contain s. This equation holds by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}.

TODO:
bei intro von natures, erklären was $\eta(\pi)(s)$ denoted

If the transition reaches s, the augmented DTMC enters the second stage.

After the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s\to s}$. Consequently, for all $s_i, s_j \in S$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(2)}, s_j^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_i, s_j) = \eta_{\text{aug}}(\pi)(s_j)$$

for every finite path π ending in t that contains s. This equation holds by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}.

Finally, the initial distribution of $\mathcal{D}^s_{aug}$ is obtained from the initial distribution of $\mathcal{D}_{s_{in}\to s}$ by assigning it to the first stage copies of the states for all $t \in S_{tr}^{\mathcal{D}_{s_{in}\to s}}$. Hence, the initial distribution agrees with that induced by η_{aug}.

Thus, $\mathcal{D}^s_{aug}$ is precisely the DTMC induced by η_{aug} on $\mathcal{I}$. Therefore,

$$\mathcal{D}^s_{aug} \models_I \mathcal{I}. \qquad \square$$

TODO:
clean up proof

Using Eq. (5.5), we can conclude that

$$Pr^{\mathcal{D}^s_{aug}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}^s_{aug}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ .$$

Thus, using the same case distinction as in Proof 12 for Lemma 3.21, we can conclude that, similar to Theorem 5.12, $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$.

TODO:
hier hinzufügen, dass EVT-via... nutzen können weil MC-aug finite ist

Conversely, if we define

$$Pr^{\mathcal{D}_{s_{in}\to s}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s\to s}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)$$

and construct η_{aug} and $\mathcal{D}^s_{aug}$, as described in this section, then $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ using the same reasoning.

Intuitively, η_{aug} is not a stationary nature, as $\eta_{aug} : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ depends not only on the current state of the model. Thus, we can conclude that there exists a nature η_{aug}, which is not stationary, for every state $s \in S^{\mathcal{I}}$, that induces a DTMC $\mathcal{D}^s_{aug}$ such that the EVT of s in $\mathcal{D}^s_{aug}$ is equal to the optimal EVT, i.e., ℓEVT or uEVT, of s.

TODO:
vllt das hier als theorem oder so

5.3 P-completeness

In this section, we further examine the complexity of calculating ℓEVTs and uEVTs by showing that the decision problem of ℓEVTs and uEVTs is P-complete.

As shown in [CHK13], computing the maximum or minimum reachability probability is P-complete for the IMDP semantic, and thus also for the UMC semantic.

TODO:
Muss ich erwähnen, dass die Berechnung zeitlich in poly(I) ist, wo I der "Input" für den IDTMC ist?

Recall that computing ℓEVTs and uEVTs is in PTIME, see Sec. 5.2.1. We now show that computing ℓEVTs and uEVTs is P-complete, by reducing it to computating minimum and maximum reachability probabilities, which is P-complete. As both ℓEVTs and uEVTs as well as minimum and maximum rachability probabilities coincide for the two semantics, it suffices to prove the following reduction for the UMC semantic.

TODO:
ist es wirklich computing, weil ich zeigs ja jz für decision problem

First, we introduce the decision problem of ℓEVTs and uEVTs as well as minimum and maximum reachability probabilities.

Definition 5.15 (Decision Problem of ℓEVTs and uEVTs)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\underline{x} \in \mathbb{R}_{>0}$, decide whether $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \underline{x}$

Similarly, given $\overline{x} \in \mathbb{R}_{>0}$, decide whether $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \overline{x}$.

TODO:
IMC I evtl in diesem chapter/section nicht mehr fixed: muss ich überprüfen für alle sections/chapter

Definition 5.16 (Decision Problem of optimal reachability probabilities)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\underline{x} \in \mathbb{R}_{>0}$, decide whether $\underline{Pr}^{\mathcal{I}}[vt_s] = \underline{x}$

Similarly, given $\overline{x} \in \mathbb{R}_{>0}$, decide whether $\overline{Pr}^{\mathcal{I}}[vt_s] = \overline{x}$.

Thus, we can now formulate the following theorem.

Theorem 5.17
The decision problem of ℓEVTs and uEVTs is P-complete.

Reduction. For the rest of this section, we fix an IDTMC $\mathcal{I}$, a set of states $T \subseteq S^{\mathcal{I}}$ and a state $s \notin S^{\mathcal{I}}$. We then construct the IDTMC $\mathcal{I}_{red} = (S^{\mathcal{I}} \cup \{s, \bot\}, \underline{\mathbf{P}}^{\mathcal{I}_{red}}, \overline{\mathbf{P}}^{\mathcal{I}_{red}}, \iota^{\mathcal{I}}_{init}, \bar{\iota}^{\mathcal{I}}_{init})$. Furthermore, we define $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \overline{\mathbf{P}}^{\mathcal{I}}(u, w)$ and $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}}(u, w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = 0$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = 1$ for $u \in T$ and $w \in S^{\mathcal{I}}$. Finally, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = 1$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = 1$.

TODO:
vllt mehr spacing und case unterscheidung statt fließtext?

Remark. Intuitively, the mapping $(\mathcal{I}, T) \mapsto (\mathcal{I}_{red})$ is computable in logarithmic space.

TODO:
Stimmt das denn auch?

Theorem 5.18
Let $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ denote a DTMC with

$$\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$$

and $\mathcal{D}_{max} \models_U \mathcal{I}$ denote a DTMC with

$$Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T) \ .$$

Then, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

Notice, that these optima exist, see the first remark in Sec. 5.2 and Def. 5.1.

TODO:
remarks zu environment machen

In other words, the optimal EVT for s in $\mathcal{I}_{red}$ is equal to the optimal reachability probability of T in $\mathcal{I}$. We prove this theorem by showing that the (maximum) reachability probabilities of T and s are identical in $\mathcal{I}_{red}$ and $\mathcal{D}_{red}$. Then, we verify that the uEVT of s is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$ and the EVT of s is equal to the reachability probability of s in $\mathcal{D}_{red}$. Lastly, we show that the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$. By combining these lemmata we then

finally prove the theorem.

Lemma 5.19
It holds that $\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the maximum reachability probabilities of T and s are identical in $\mathcal{I}_{red}$.

Proof 14
As $\mathbf{P}^{\mathcal{D}}(t,s) > 0$ if and only if $t \in T$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, it follows from Lemma 3.7 that $Pr^{\mathcal{D}}(\Diamond s) = \sum_{t \in T} \mathbf{P}(t,s) \cdot Pr^{\mathcal{D}}(\Diamond T)$. Because, $\mathbf{P}^{\mathcal{D}}(t,s) = 1$ for all $t \in T$, we can conclude that $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond s)$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$.

TODO:
do i need to show this "follows from lemma 3.11" more thoroughly?

Lemma 5.20
It holds that $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the uEVT of $s \in S^{\mathcal{I}_{red}}$ is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$.

Proof 15
$\mathbb{E}^{\mathcal{D}}[vt_s] = Pr(\Diamond s) \cdot \mathbb{E}_s^{\mathcal{D}}[vt_s]$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, see **??**. As, $\overline{Pr}^{\mathcal{I}_{red}}(s,t) > 0$ if and only if $t = \tau$ and τ is absorbing, it follows that $\mathbb{E}_s^{\mathcal{D}}[vt_s] = 1$, as $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 0$, which concludes the proof. □

Lemma 5.21
It holds that $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

In other words, the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$.

Proof 16
Firstly, if $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ (as defined in the construction), then $Pr^{\mathcal{D}_{red}}(\Diamond s) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$, see Lemma 5.20. Moreover, using Lemma 5.19, we can then conclude that $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$.

TODO:
Sind diese Beweischritte und andere ähnliche verständlich oder sind die zu grob und müssen durch kurze Erläuterungen besser erklärt werden?

As $Pr^{\mathcal{D}_{red}}(\Diamond T)$ and $Pr^{\mathcal{D}_{max}}(\Diamond T)$ are only dependent on $\mathbf{P}^{\mathcal{D}_{red}}(u,w)$ and $\mathbf{P}^{\mathcal{D}}(u,w)$

with $u \in S^{\mathcal{I}} \backslash T$ respectively, see Lemma 3.7, and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \overline{\mathbf{P}}^{\mathcal{I}}(u,w)$ as well as $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \underline{\mathbf{P}}^{\mathcal{I}}(u,w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$, we can conclude that $\overline{Pr}^{\mathcal{I}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$. By the definition of our construction, $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$ (see first paragraph of this proof) which concludes the proof.

TODO:
Muss zum Beispiel auch dieser letzte Satz aufführlicher formuliert werden, also quasi die Werte nochmal einsetzen so dass man die finale Form (des Lemma) erhält?

Finally, we can now prove Theorem 5.18.

Proof 17
We simply need to combine the results of the lemmata. According to Lemma 5.21 $Pr^{\mathcal{D}_{max}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond T)$. Furthermore, using Lemma 5.19, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. Finally, Lemma 5.20 shows that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. Thus, combining these results $Pr^{\mathcal{D}_{max}}(\Diamond T) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. By definition $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Therefore, the given maximum reachability probability $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ of any IDTMC $\mathcal{I}$ can be determined through calculating uEVTs.□

Conversely, the same holds using this reasoning for the ℓEVTs.

Thus, as we can calculate the reachabilty minimum and maximum reachability probabilities of any given $T \subseteq S^{\mathcal{I}}$ using ℓEVTs and uEVTs, we can decide the problem of optimal reachability probabilities using $\underline{x} = \underline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$ and $\overline{x} = \overline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$. As the decision problem of optimal reachability probabilities is P-complete, it thus follows that the decision problem of ℓEVTs and uEVTs is P-complete as well.

Chapter 6

Stationary Distributions in IDTMCs

In this chapter, we examine how ℓEVTs and uEVTs can be used to calculate stationary distributions in IDTMCs. For this we start by .... To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$ with $\mathcal{D} \models_U \mathcal{I}$.

TODO:
fix DTMC and IDTMC

We define stationary distributions, also called long-run distributions, as done in [BK08].

Definition 6.1 (Stationary Distributions in DTMCs)
Given a finite DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ be a finite Markov chain and $s \in S$. The *long-run average probability* $\theta^{\mathcal{D}}(s)$ is given by

$$\theta^{\mathcal{D}}(s) = \lim_{n \to \infty} \frac{1}{n} \cdot \sum_{i=1}^{n} Pr(\Diamond^{=k} s) \ .$$

In other words, the stationary distribution $\Theta^{\mathcal{D}}(t)$ describes the probability that $\mathcal{D}$ is in state t after k steps, as k tends towards infinity. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $\theta^{\mathcal{D}}$.

Similarly, we now define stationary distributions for IDTMCs. First, recall that the DTMCs $\mathcal{D}_\eta \models_I \mathcal{I}$ that are induced by some nature η can be infinite, i.e., $|\mathcal{D}_\eta| = \infty$, see Def. 4.7. Intuitively, as every state $s \in S^{\mathcal{D}_{eta}}$ in these DTMCs is only visited once, $\theta(s) = 0$. In other words, there exists no stationary distribution for these states.

TODO:
noch mehr intuition mittels der T x theta = theta formel

Thus, we restrict this examination of BSCCs, in this thesis, to the UMC semantic.

Definition 6.2 (Stationary Distributions in IDTMCs)
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, we define

$$\Theta(s) = \{\theta(s)^{\mathcal{D}} \mid \mathcal{D} \models_U \mathcal{I}\}$$

as the set of stationary distributions of s on $\mathcal{I}$.
Furthermore, we define $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ such that

$$\underline{\theta}^{\mathcal{I}}(s) = \inf_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \qquad \text{and} \qquad \overline{\theta}^{\mathcal{I}}(s) = \sup_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \ .$$

In other words, $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ describe the minimum and maximum stationary distribution of s among the DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $\underline{\theta}^{\mathcal{I}}$ and $\overline{\theta}^{\mathcal{I}}$.

Moreover, BSCCs are defined as follows for IDTMCs.

Definition 6.3 (BSCCs for IDTMCs)
Let $\mathcal{I}$ be an IDTMC, then $B^{\mathcal{I}} \subseteq S^{\mathcal{I}}$ is a BSCC, if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $B^{\mathcal{I}}$ is a BSCC in the DTMC $\mathcal{D}$, see Def. 3.9.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $B^{\mathcal{I}}$.

In contrast to EVTs, when considering the stationary distribution of a DTMC or IDTMC, we are interested in the recurrent, not transient, states.

TODO:
noch etwas mehr intuition geben

Recall, that the recurrent states of a DTMC $\mathcal{D}$ are precisely the states contained in the BSCCs of $\mathcal{D}$, see Sec. 3.1. Thus, by collapsing each BSCCS of $\mathcal{D}$ into a single state, the stationary distribution $\theta^{\mathcal{D}}$ of the recurrent states of a DTMC $\mathcal{D}$ is characterized by

$$\theta^{\mathcal{D}}(s) = Pr^{\mathcal{D}}(\Diamond B) \cdot \theta^{\mathcal{D}|_B}(s) \ , \tag{6.1}$$

for all $s \in B$, as proven in [Kul16], where $\theta^{\mathcal{D}|_B}$ denotes the stationary distribution of $\mathcal{D}$ restricted to B. Moreover, in [MKQW25], it is shown that the stationary distribution within a BSCC can be computed through its EVTs. For this they construct a DTMC $\mathcal{B}^{\hat{v}}$ for every BSCC B with any arbitrary state $v \in S^{\mathcal{D}|_B}$ in which all incoming

transitions of v are redirected to the fresh absorbing state $\hat{v}$. As v is arbitrary, we fix v and $\hat{v}$ for the remainder of this chapter.

TODO:
vllt reicht das nicht: muss nochmal genauer argumentieren, weshalb auch für IDTMC v arbiträr

Construction. More precisely, $\mathcal{B}\,{\restriction^{\hat{v}}} = (S^B \cup \{\hat{v}\}, \mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}, \iota_{init}^{\mathcal{B}\restriction^{\hat{v}}})$, where S^B denotes the states that are contained in B. Moreover, $\mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}(\hat{v},\hat{v}) = 1$, $\mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}(s,\hat{v}) = \mathbf{P}^{\mathcal{D}}(s,v)$, and $\mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}(s,v) = 0$ for all $s \in S^{\mathcal{D}|_B}$. Lastly, $\mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}(s,t) = \mathbf{P}^{\mathcal{D}}(s,t)$ for all $s \in S$, $t \in S\backslash\{v\}$, and $\iota_{init}^{\mathcal{B}\restriction^{\hat{v}}}(v) = 1$.

The stationary distribution within a DTMC is then given by

$$\theta^{\mathcal{D}|_B}(s) = \frac{\mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_s]}{\sum_{t\in S}\mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_t]}\,, \tag{6.2}$$

as proven in [MKQW25].

Similarly to the DTMC $\mathcal{B}\restriction^{\hat{v}}$, we now define the IDTMC $\mathcal{I}\restriction^{\hat{v}}$ as follows.

Construction. $\mathcal{I}\,{\restriction^{\hat{v}}} = (S^B \cup \{\hat{v}\}, \underline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}, \overline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}, \underline{\iota}_{init}^{\mathcal{B}\restriction^{\hat{v}}}, \overline{\iota}_{init}^{\mathcal{B}\restriction^{\hat{v}}})$, where S^B denotes the states that are contained in B. Moreover, $\overline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(\hat{v},\hat{v}) = \underline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(\hat{v},\hat{v}) = 1$, $\overline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(s,\hat{v}) = \overline{\mathbf{P}}^{\mathcal{I}}(s,v)$, $\underline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(s,\hat{v}) = \underline{\mathbf{P}}^{\mathcal{I}}(s,v)$, and $\overline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(s,v) = \underline{\mathbf{P}}^{\mathcal{B}\restriction^{\hat{v}}}(s,v) = 0$ for all $s \in S^{\mathcal{D}|_B}$. Lastly, $\mathbf{P}^{\mathcal{B}\restriction^{\hat{v}}}(s,t) = \mathbf{P}(s,t)$ for all $s \in S$, $t \in S\backslash\{v\}$, and $\iota_{init}^{\mathcal{B}\restriction^{\hat{v}}}(v) = 1$.

Intuitively, as $\mathcal{D} \models_U \mathcal{I}$ it follows that, by construction, $\mathcal{B}\restriction^{\hat{v}} \models_U \mathcal{I}\restriction^{\hat{v}}$.

Assumption 2
In the rest of this Chapter, we assume that all IDTMCs $\mathcal{I}$ are graph preserving, meaning that there exists no lower transition probability function such that $\underline{\mathbf{P}}^{\mathcal{I}}(s,t) = 0$ for any $s,t \in S^{\mathcal{I}}$. This implies that there exists no $\mathcal{D} \models_U \mathcal{I}$ such that some $s \in S^{\mathcal{I}}$ is not reachable in $\mathcal{D}$.

Given this assumption, we can prove the following theorem.

Theorem 6.4
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}_{re}$, and any pair of BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'} \subseteq S^{\mathcal{I}}$ with $B \neq B'$, it holds that $s \in B$ implies $s \notin B'$.

In other words, every state $S^{\mathcal{I}}_{re}$ corresponds to precisely one BSCC in $\mathcal{I}$.

Proof 18
Assume that for some IDTMC $\mathcal{I}$ there exists a state s, two BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'}$ with $B \not= B'^{\mathcal{D}'}$, and $\mathcal{D}, \mathcal{D}' \models \mathcal{I}$, such that $s \in B^{\mathcal{D}}$ and $s \in B'^{\mathcal{D}'}$. As $B^{\mathcal{D}} \neq B'^{\mathcal{D}'}$, there

exists at least one state $t \in S^{\mathcal{I}}$ that is only contained in either $B^{\mathcal{D}}$ or $B'^{\mathcal{D}'}$, i.e., $t \in (B \backslash B') \cup (B' \backslash B)$.

Without loss of generality, let us assume $t \in B^{\mathcal{D}}$. Hence, by Def. 3.9, there exists a path from s to t in $\mathcal{D}$. However, as noted the graph relations for all satisfying DTMCs of graph preserving IDTMCs are identical. Thus, there must also exist a path from s to t in $\mathcal{D}'$. However, $t \notin B'^{\mathcal{D}'}$, which contradicts Def. 3.9. □

TODO:
BSCC mit rücksicht auf IMC definieren, also dass BSSC von DTMC, die IDTMC satisfied

Thus, we can conclude for graph preserving IDTMCs that any state $s \in S^{\mathcal{I}}$ belongs only to a singular BSCC B, i.e., the BSCCs for all satisfying DTMCs are identical. This allows us, to optimize the reachability of B ($Pr^{\mathcal{D}}(\Diamond B)$) and the stationary distribution restricted to B ($\theta^{\mathcal{D}|_B}$) of Eq. (6.1) independently as shown in the following.

Therefore, we can intuitively calculate the stationary distributions for a state $s \in \mathcal{I}$ by solving following NLP:

$$\begin{aligned} \text{opt} \quad & \frac{\mathbb{E}^{\mathcal{B}^{\Cap \hat{v}}}[vt_s]}{\sum_{t \in S^{\mathcal{B}^{\Cap \hat{v}}}} \mathbb{E}^{\mathcal{B}^{\Cap \hat{v}}}[vt_t]} \quad \text{opt} \in \{\max, \min\} \\ \text{s.t.} \quad & \mathcal{B}^{\Cap \hat{v}} \models_U \mathcal{I}^{\Cap \hat{v}} \quad . \end{aligned}$$

Theorem 6.5
Given an IDTMC $\mathcal{I}$, a state s, the corresponding BSCC $B^{\mathcal{I}}$ with $s \in B^{\mathcal{I}}$, and the corresponding $\mathcal{I}^{\Cap \hat{v}}$, let $\overline{\theta}^{\mathcal{I}|_B}$ and $\underline{\theta}^{\mathcal{I}|_B}$ be the the solutions of the NLP for $\mathcal{I}^{\Cap \hat{v}}$ when maximizing and minimizing, respectively. We can then calculate the minimum and maximum stationary distribution using

$$\underline{\theta}^{\mathcal{I}}(s) = \underline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \underline{\theta}^{\mathcal{I}|_B}$$

and

$$\overline{\theta}^{\mathcal{I}}(s) = \overline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \overline{\theta}^{\mathcal{I}|_B}$$

similar to Eq. (6.1) using Eq. (6.2).

We provide a short proof by contradiction of this theorem

Proof 19
Let us assume that there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\overline{\theta}^{\mathcal{I}}(s) < \theta^{\mathcal{D}}(s)$. Using

Theorem 6.5 and Eq. (6.1), we can thus conclude that either

$$Pr^{\mathcal{D}}(\Diamond B) > \overline{Pr}^{\mathcal{I}}(\Diamond B) \qquad \text{or} \qquad \frac{\mathbb{E}^{\mathcal{B}^{\hat{\upsilon}}}[vt_s]}{\sum_{t \in S} \mathbb{E}^{\mathcal{B}^{\hat{\upsilon}}}[vt_t]} > \overline{\theta}^{\mathcal{I}|_B} \ .$$

However, as $\mathcal{D} \models_U \mathcal{I}$ and $\mathcal{B}^{\hat{\upsilon}} \models_U \mathcal{I}^{\hat{\upsilon}}$, by the definition of $\overline{Pr}^{\mathcal{I}}$ and the maximality of $\overline{\theta}^{\mathcal{I}|_B}$ as a solution of the NLP, this leads to a contradiciton. Furthermore, as s belongs to a singular BSCC B, every $\mathcal{B}^{\hat{\upsilon}} \models_U \mathcal{I}^{\hat{\upsilon}}$ is necessarily constructed from B. Thus, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$, such that $Pr^{\mathcal{D}}(\Diamond B) = \overline{Pr}^{\mathcal{I}}(\Diamond B)$ and $\theta^{\mathcal{D}|_B} = \overline{\theta}^{\mathcal{I}|_B}$. The proof for $\underline{\theta}^{\mathcal{I}}(s)$ follows analogously by inverting the inequalities.□

Chapter 7

Conclusion

Appendix A

Appendix

Enter text

Bibliography

[ADD00] Robert B. Ash and Catherine C. Doleans-Dade. *Probability and Measure Theory*. Elsevier Science, 2000.

[Afr20] Feri Afrinaldi. Exploring product lifecycle using Markov chain. *Procedia Manufacturing*, 43:391–398, 2020. Sustainable Manufacturing - Hand in Hand to Sustainability on Globe: Proceedings of the 17th Global Conference on Sustainable Manufacturing.

[AGM95] Samson Abramsky, Dov M. Gabbay, and Thomas S. E. Maibaum, editors. *Handbook of Logic in Computer Science*, volume 3. Oxford University Press, USA, 1995.

[AN14] Subil Abraham and Suku Nair. Cyber Security Analytics: A Stochastic Model for Security Quantification Using Absorbing Markov Chains. *Journal of Communications*, pages 899–907, 12 2014.

[Bad] Thom Badings. Robust Verification of Stochastic Systems: Guarantees in the Presence of Uncertainty.

[BDL+17] Anicet Bart, Benoit Delahaye, Didier Lime, Eric Monfroy, and Charlotte Truchet. Reachability in Parametric Interval Markov Chains using Constraints, 2017.

[BDLM21] Giovanni Bacci, Benoît Delahaye, Kim G. Larsen, and Anders Mariegaard. *Quantitative Analysis of Interval Markov Chains*, pages 57–77. Springer International Publishing, Cham, 2021.

[BK08] Christal Baier and {Joost P.} Katoen. *Principles of Model Checking*. MIT Press, United States, May 2008.

[BKL+17] Christel Baier, Joachim Klein, Linda Leuschner, David Parker, and Sascha Wunderlich. Ensuring the Reliability of Your Model Checker: Interval Iteration for Markov Decision Processes. In Rupak Majumdar and Viktor Kunčak, editors, *Computer Aided Verification*, pages 160–180, Cham, 2017. Springer International Publishing.

[BLNW14] Fabrizio Biondi, Axel Legay, Bo Friis Nielsen, and Andrzej Wąsowski. Maximizing entropy over Markov processes. *Journal of Logical and Algebraic Methods in Programming*, 83(5):384–399, 2014. The 24th Nordic Workshop on Programming Theory (NWPT 2012).

[BLW13] Michael Benedikt, Rastislav Lenhardt, and James Worrell. LTL Model Checking of Interval Markov Chains. In Nir Piterman and Scott A. Smolka, editors, *Tools and Algorithms for the Construction and Analysis of Systems*, pages 32–46, Berlin, Heidelberg, 2013. Springer Berlin Heidelberg.

[Cas09] Hal Caswell. Stage, age and individual stochasticity in demography. *Oikos*, 118(12):1763–1782, 2009.

[CHK13] Taolue Chen, Tingting Han, and Marta Kwiatkowska. On the complexity of model checking interval-valued discrete time Markov chains. *Information Processing Letters*, 113(7):210–216, 2013.

[CSH08] Krishnendu Chatterjee, Koushik Sen, and Thomas A. Henzinger. Model-Checking ω-Regular Properties of Interval Markov Chains. In Roberto Amadio, editor, *Foundations of Software Science and Computational Structures*, pages 302–317, Berlin, Heidelberg, 2008. Springer Berlin Heidelberg.

[FJ90] Noah E. Friedkin and Eugene C. Johnsen. Social influence and opinions. *The Journal of Mathematical Sociology*, 15(3-4):193–206, 1990.

[FRG13] Lance Fiondella, Sanguthevar Rajasekaran, and Swapna S. Gokhale. Efficient Software Reliability Analysis With Correlated Component Failures. *IEEE Transactions on Reliability*, 62(1):244–255, 2013.

[Gal17] Serge Galam. The Trump phenomenon: An explanation from sociophysics. *International Journal of Modern Physics B*, 31(10):1742015, April 2017.

[GLD00] Robert Givan, Sonia Leach, and Thomas Dean. Bounded-parameter Markov decision processes. *Artificial Intelligence*, 122(1):71–109, 2000.

[Gok04] Swapna S. Gokhale. Quantifying the Variance in Application Reliability. In *Proceedings of the 10th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC'04)*, PRDC '04, page 113–121, USA, 2004. IEEE Computer Society.

[GT02] S.S. Gokhale and K.S. Trivedi. Reliability prediction and sensitivity analysis based on software architecture. In *13th International Symposium on Software Reliability Engineering, 2002. Proceedings.*, pages 64–75, 2002.

[GWHT04] Swapna S. Gokhale, W. Eric Wong, J.R. Horgan, and Kishor S. Trivedi. An analytical approach to architecture-based software performance and reliability prediction. *Performance Evaluation*, 58(4):391–412, 2004.

[HBK17] Lisa Hutschenreiter, Christel Baier, and Joachim Klein. Parametric Markov Chains: PCTL Complexity and Fraction-free Gaussian Elimination. *Electronic Proceedings in Theoretical Computer Science*, 256:16–30, 2017.

[JJK22] Nils Jansen, Sebastian Junges, and Joost-Pieter Katoen. Parameter Synthesis in Markov Models: A Gentle Survey, 2022.

[JL91] B. Jonsson and K.G. Larsen. Specification and refinement of probabilistic processes. In *[1991] Proceedings Sixth Annual IEEE Symposium on Logic in Computer Science*, pages 266–277, 1991.

[Kat16] Joost-Pieter Katoen. The Probabilistic Model Checking Landscape. In *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science*, LICS '16, page 31–45, New York, NY, USA, 2016. Association for Computing Machinery.

[KS76] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Undergraduate Texts in Mathematics. Springer, 1976.

[KU02] Igor O. Kozine and Lev V. Utkin. Interval-Valued Finite Markov Chains. *Reliable Computing*, 8(2):97–113, 2002.

[Kul16] Vidyadhar Kulkarni. *Modeling and analysis of stochastic systems: Third edition*. 11 2016.

[Kuz91] V. P. Kuznetsov. *Interval Statistical Models*. Radio and Communication, 1991.

[LY08] David G. Luenberger and Yinyu Ye. *Linear and Nonlinear Programming*, volume 228 of *International Series in Operations Research & Management Science*. Springer, 3rd edition, 2008.

[MAK06] Yasunari Matsuno, Yoshihiro Adachi, and Yasushi Kondo. Application of Markov Chain Model to Calculate the Average Number of Times of Use of a Material in Society. An Allocation Methodology for Open-Loop Recycling. Part 1: Methodology Development. *The International Journal of Life Cycle Assessment*, 11(5):354–360, 2006.

[Mar06] Andrey Andreyevich Markov. Extension of the law of large numbers to dependent quantities. *Izv. Fiz.-Matem. Obsch. Kazan Univ.(2nd Ser)*, 15(1):135–156, 1906.

[MKQW25] Hannah Mertens, Joost-Pieter Katoen, Tim Quatmann, and Tobias Winkler. Computing Expected Visiting Times and Stationary Distributions in Markov Chains: Fast and Accurate. *Journal of Automated Reasoning*, 69(3), August 2025.

[MMAG11] Indika Meedeniya, Irene Moser, Aldeida Aleti, and Lars Grunske. Architecture-based reliability evaluation under uncertainty. In *Proceedings of the Joint ACM SIGSOFT Conference – QoSA and ACM SIGSOFT Symposium – ISARCS on Quality of Software Architectures – QoSA and Architecting Critical Systems – ISARCS*, QoSA-ISARCS '11, page 85–94, New York, NY, USA, 2011. Association for Computing Machinery.

[Neu77] C. Neumann. *Untersuchungen über das logarithmische und Newton'sche Potential*. B. G. Teubner, 1877.

[OA14] Hamed Orojloo and Mohammad Abdollahi Azgomi. A method for modeling and evaluation of the security of cyber-physical systems. In *2014 11th International ISC Conference on Information Security and Cryptology*, pages 131–136, 2014.

[PPH13] Mahendra Piraveenan, Mikhail Prokopenko, and Liaquat Hossain. Percolation Centrality: Quantifying Graph-Theoretic Impact of Nodes during Percolation in Networks. *PLOS ONE*, 8(1):1–14, 01 2013.

[PRT10] R. Pietrantuono, S. Russo, and K. S. Trivedi. Online Monitoring of Software System Reliability. In *2010 European Dependable Computing Conference*, pages 209–218, 2010.

[RC18] Gregory Roth and Hal Caswell. Occupancy time in sets of states for demographic models. *Theoretical Population Biology*, 120:62–77, 2018.

[SJT05] Vibhu Saujanya Sharma, Pankaj Jalote, and Kishor S. Trivedi. Evaluating Performance Attributes of Layered Software Architecture. In George T. Heineman, Ivica Crnkovic, Heinz W. Schmidt, Judith A. Stafford, Clemens Szyperski, and Kurt Wallnau, editors, *Component-Based Software Engineering*, pages 66–81, Berlin, Heidelberg, 2005. Springer Berlin Heidelberg.

[Spr18] Jeremy Sproston. *Qualitative Reachability for Open Interval Markov Chains*, page 146–160. Springer International Publishing, 2018.

[ST06] Vibhu Saujanya Sharma and Kishor S. Trivedi. Reliability and Performance of Component Based Software Systems with Restarts, Retries, Reboots and Repairs. In *2006 17th International Symposium on Software Reliability Engineering*, pages 299–310, 2006.

[SVA06] Koushik Sen, Mahesh Viswanathan, and Gul Agha. Model-Checking Markov Chains in the Presence of Uncertainties. In Holger Hermanns and Jens Palsberg, editors, *Tools and Algorithms for the Construction and Analysis of Systems*, pages 394–410, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.

[TB07] Ambuj Tewari and Peter L. Bartlett. Bounded parameter Markov decision processes with average reward criterion. In *Proceedings of the 20th*

Annual Conference on Learning Theory, COLT'07, page 263–277, Berlin, Heidelberg, 2007. Springer-Verlag.

[Teo13] Mingmei Teo. Interval Markov Chains: Performance Measures and Sensitivity Analysis. Master of philosophy thesis, The University of Adelaide, Adelaide, Australia, 2013.

[Wal96] Peter Walley. Measures of uncertainty in expert systems. *Artificial Intelligence*, 83(1):1–58, 1996.